

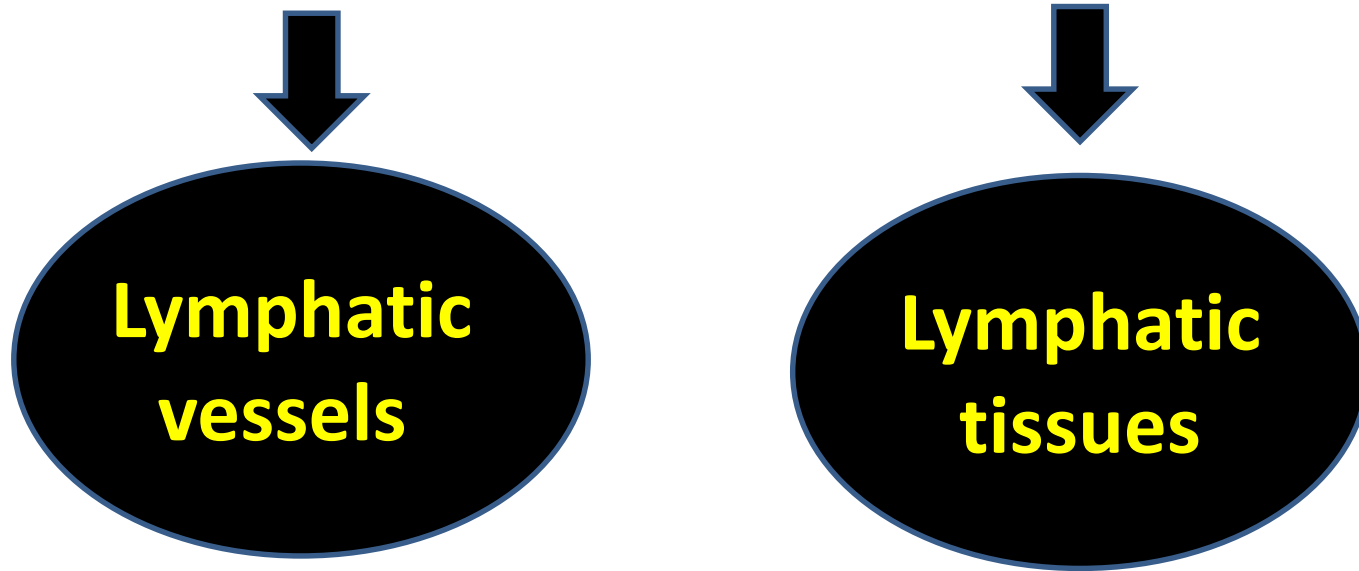
Immune system (Lymphatic system)

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Immune system (Lymphatic system)

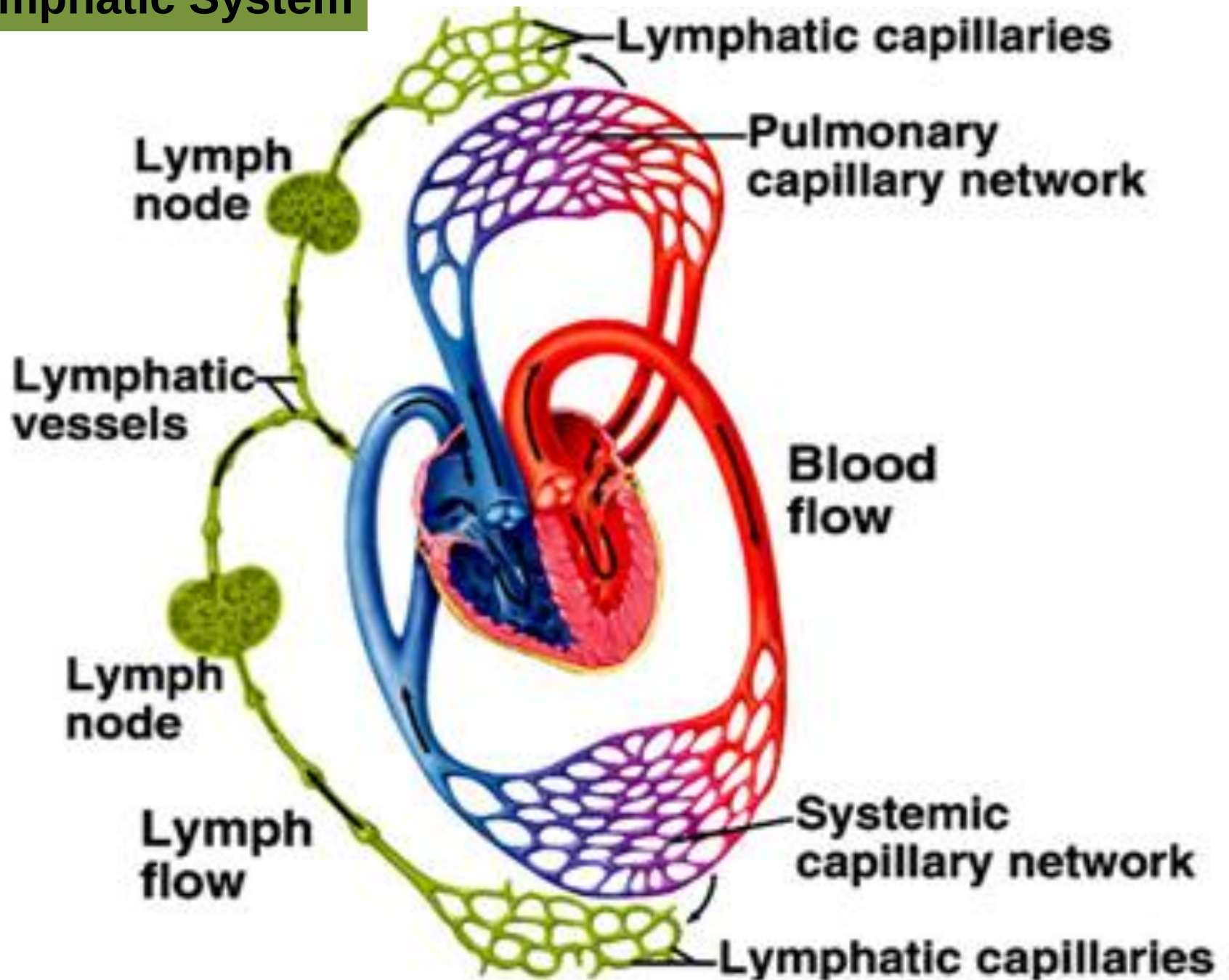
- Distinguish self molecules from non self (foreign) molecules.
- Neutralize or inactivate foreign molecules (viruses, bacteria, and parasites).
- Destroy microorganisms or other cells (such as virus-infected cells, cells of transplanted organs, and cancer cells).
- On occasion, the immune system of an individual reacts against its own normal body tissues, causing autoimmune disease.

Immune system (Lymphatic system)

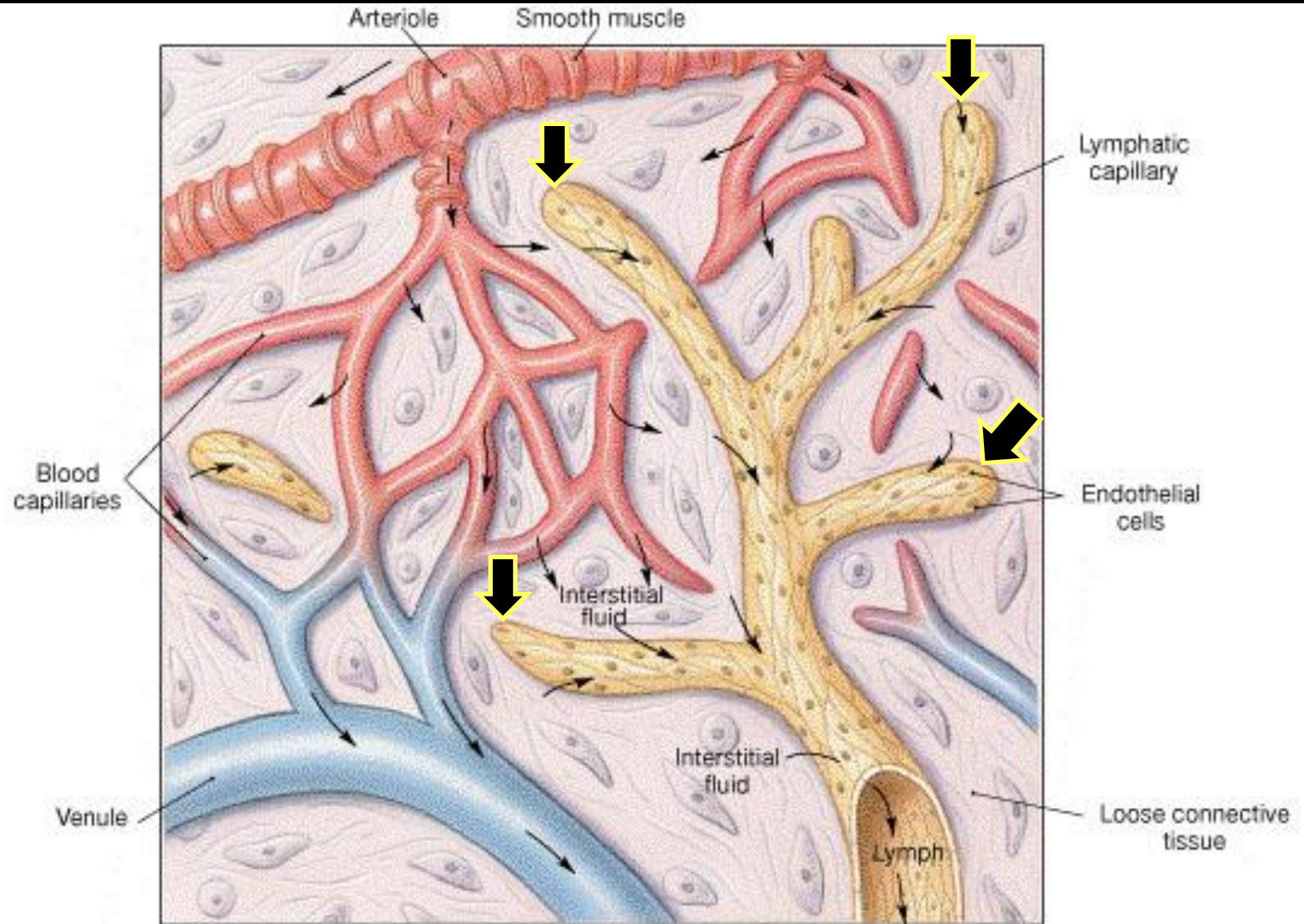


- Lymphatic vessels are blind end tubes.
- Assist the cardiovascular system in removal of tissue fluid from interstitial tissue and carry this fluid to the blood.

Lymphatic System



Lymphatic vessels



(a) Association of blood capillaries, tissue, and lymphatic capillaries

Cells of the Immune System

lymphocytes

**plasma
cells**

**mast
cells**

neutrophils

eosinophils

**cells of the
mononuclear
phagocyte
system.**

Lymphocytes

T Lymphocytes (T Cells)

Cell mediated
Immunity

T Killer

T Helper

T suppressor

B Lymphocytes (B Cells)

Humeral
Immunity

Plasma Cells

Antibodies

Natural Killer cells (NK cells)

Innate
Immunity

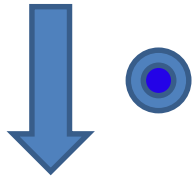
Destroy

Tumor
cells

Large
Viruses

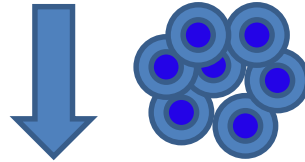
Lymphoid Tissue

The cells of the immune system (particularly lymphocytes) are scattered through out the body either as:



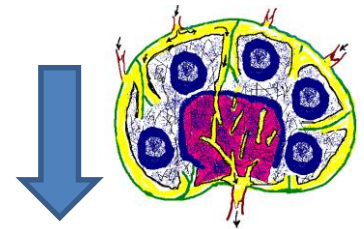
Isolated cells

Free in the
connective
tissue



**Non encapsulated
Aggregations
(lymph nodules)**

Beneath
mucosal
surfaces
mucosa
associated
lymphoid tissue
(MALT)



**Within
lymphoid organs**

Thymus, lymph
nodes, spleen
and the hemal
nodes

Lymphoid nodules (lymphoid follicles)

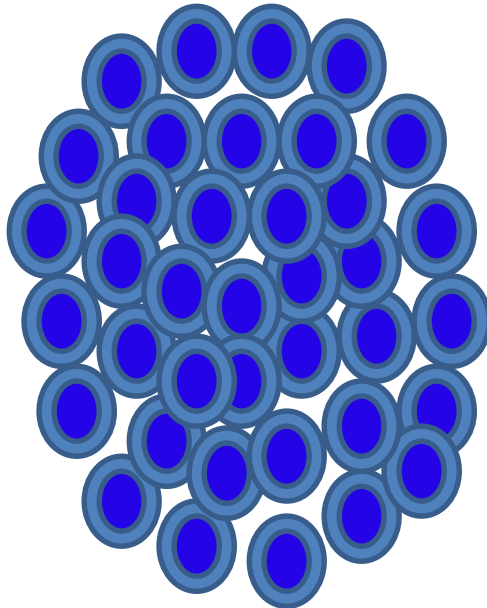
primary lymphoid nodules

Antigen



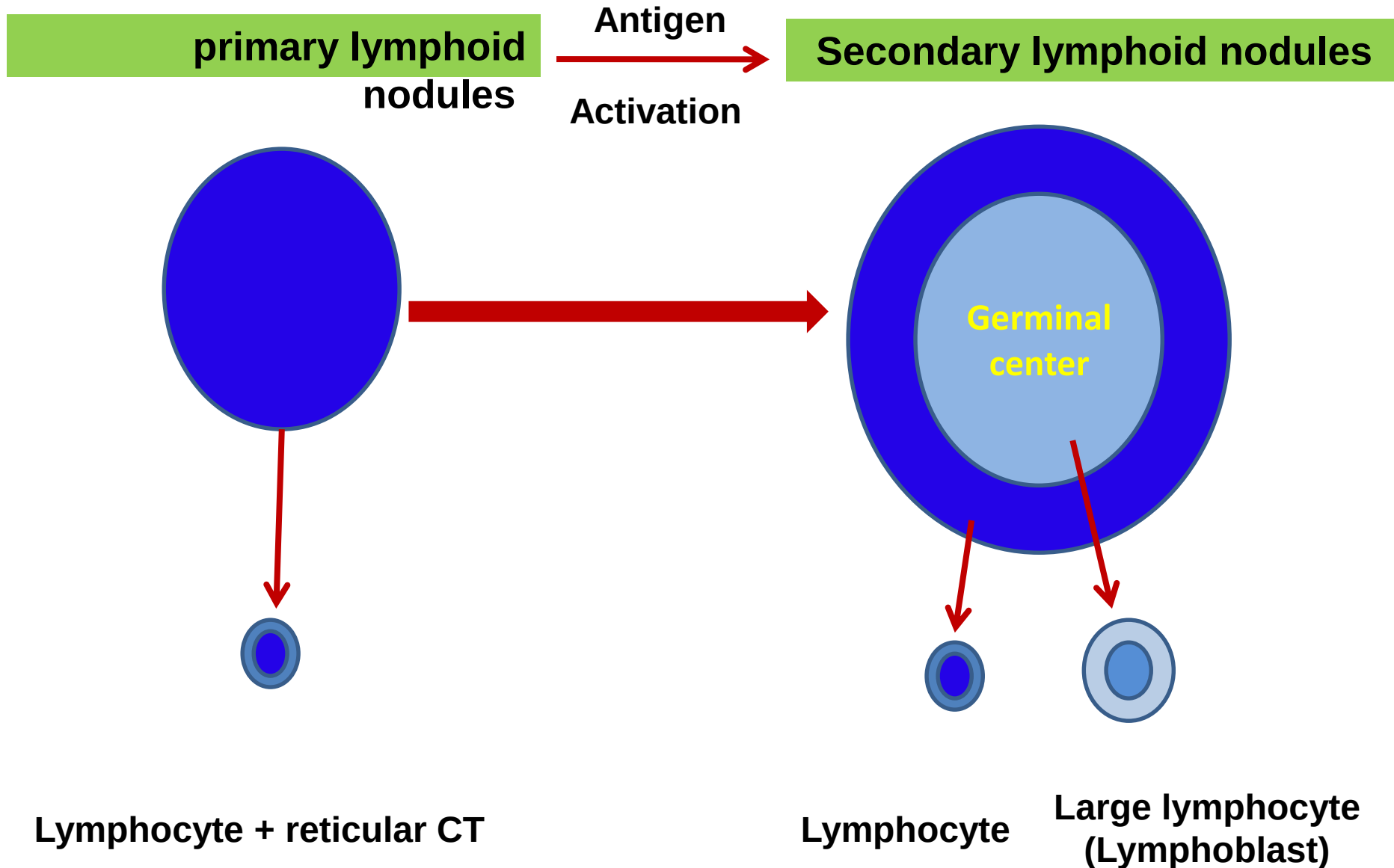
Activation

Secondary lymphoid nodules



Lymphocyte + reticular CT

Lymphoid nodules (lymphoid follicles)



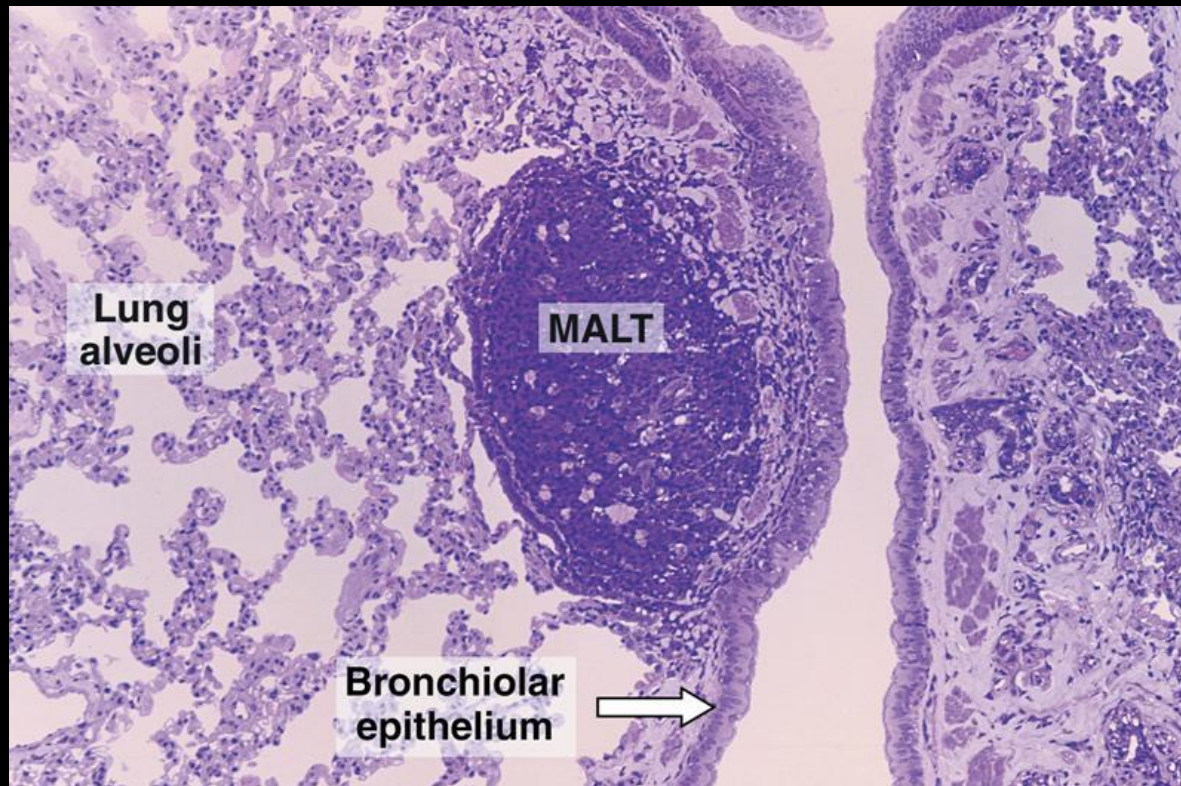
Lymphoid nodules

(lymphoid follicles)

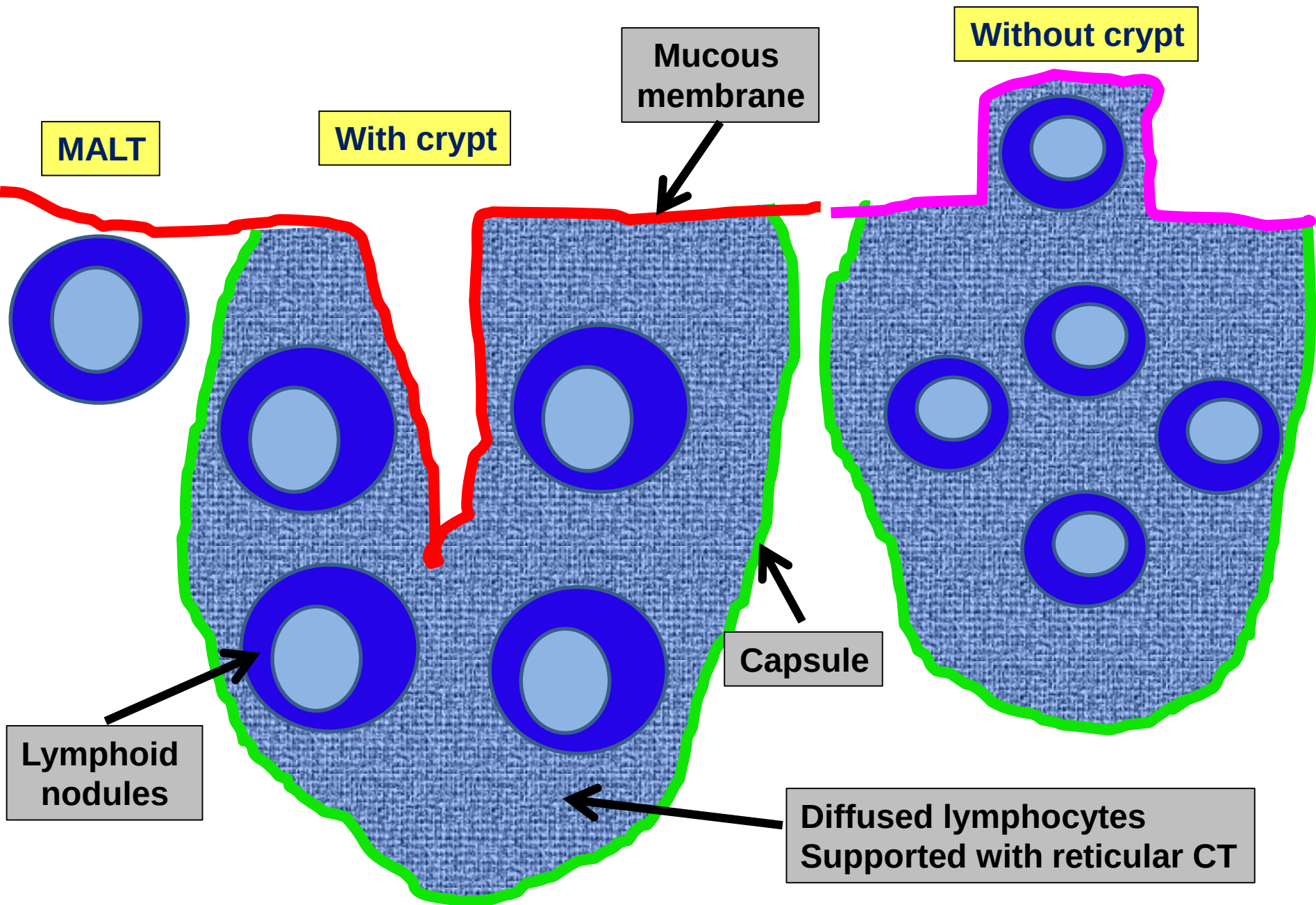
- Groups of lymphocytes (Mainly B lymphocytes). arranged as spheres.
- All lymphoid tissues are supported with **reticular CT except in the thymus.**
- Activation of **primary lymphoid nodules** (recognition of antigens), leads to proliferation of lymphocytes in the center of the nodule, which then stains lighter and is called a **germinal center** leading to formation of **secondary lymphoid nodule.**
- After completion of the immune response, the germinal center may disappear.
- The germinal centers contain a special cell, **the follicular dendritic cell**, that has many processes that bind antigen on their surfaces, to be presented to B lymphocytes.

Mucosa-Associated Lymphoid Tissue (MALT)

The mucosa and submucosa of the digestive, respiratory, and genitourinary tracts contain a large amount of diffuse collections of lymphocytes, IgA-secreting plasma cells, APCs, and lymphoid nodules (MALT) to protect them from foreign antigens.



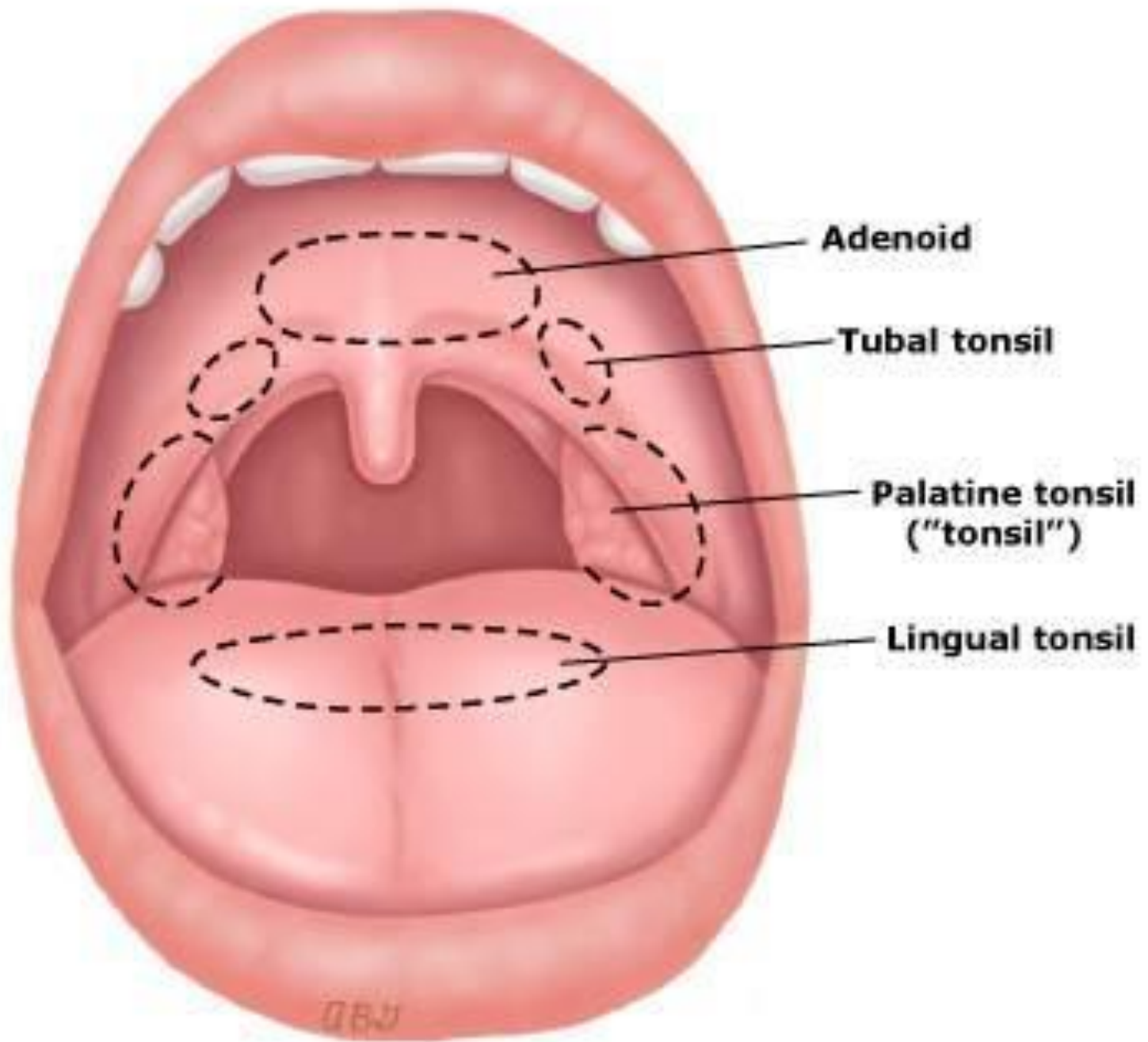
Tonsils



Tonsils

- Tonsils belong to the **MALT**.
- As they are **incompletely encapsulated**, they are **considered organ**.
- Tonsils are lymphoid tissue **lies beneath**, and in contact with, **the epithelium of the initial portion of the digestive and respiratory tracts**.
- Depending on their location, tonsils in the mouth and pharynx are called **palatine, pharyngeal**, or **lingual**.

Tonsils



Tonsils are classified according to the presence or absence of crypts (deep invaginations) into:

Tonsils with crypt

Palatine tonsils of :
human, ruminants & horse.

Lingual tonsils of :
human & most domestic animals.

Tonsils without crypt

Pharyngeal tonsil of:
human & most domestic animals.

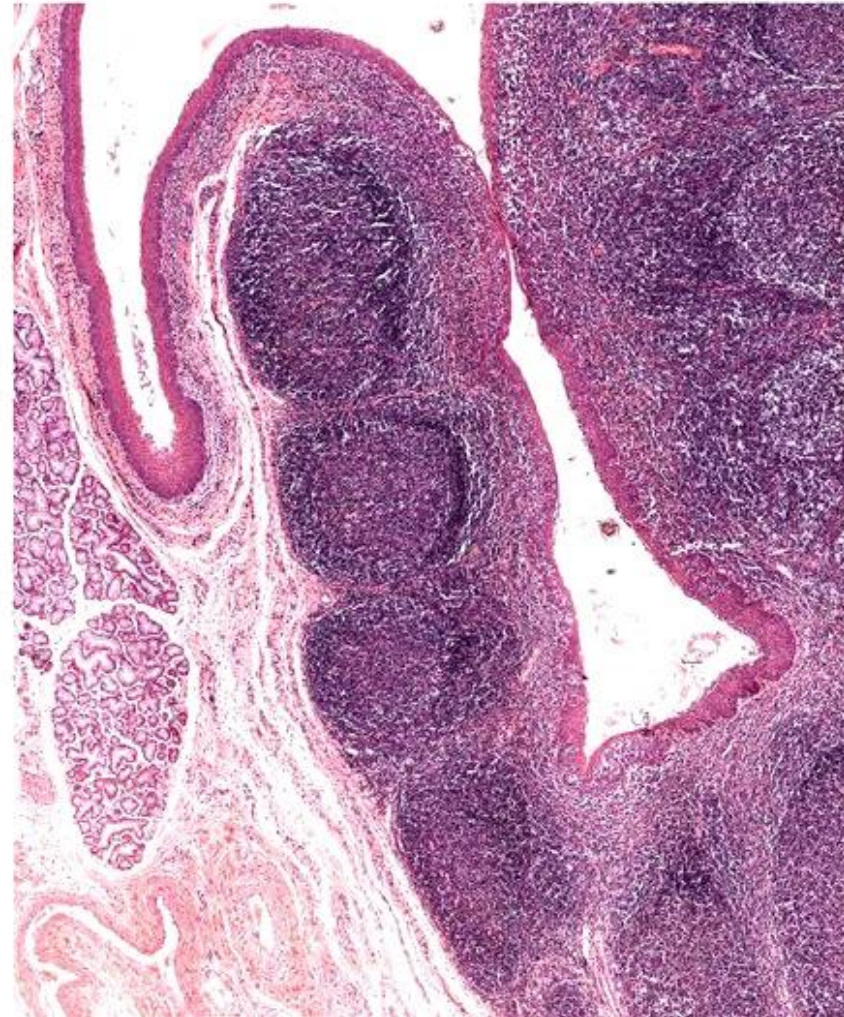
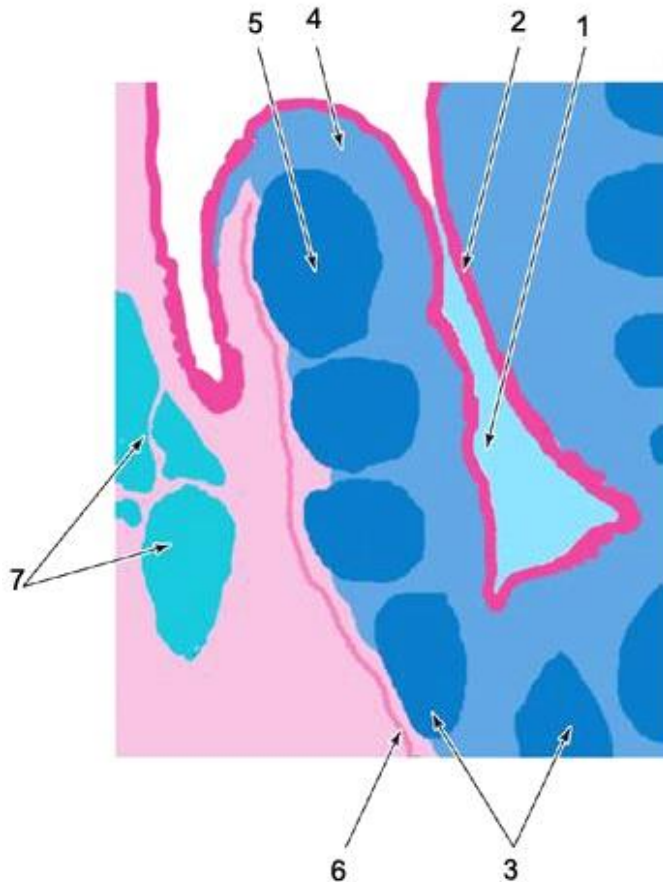
Palatine tonsils of:
Dog & cat

Palatine Tonsils

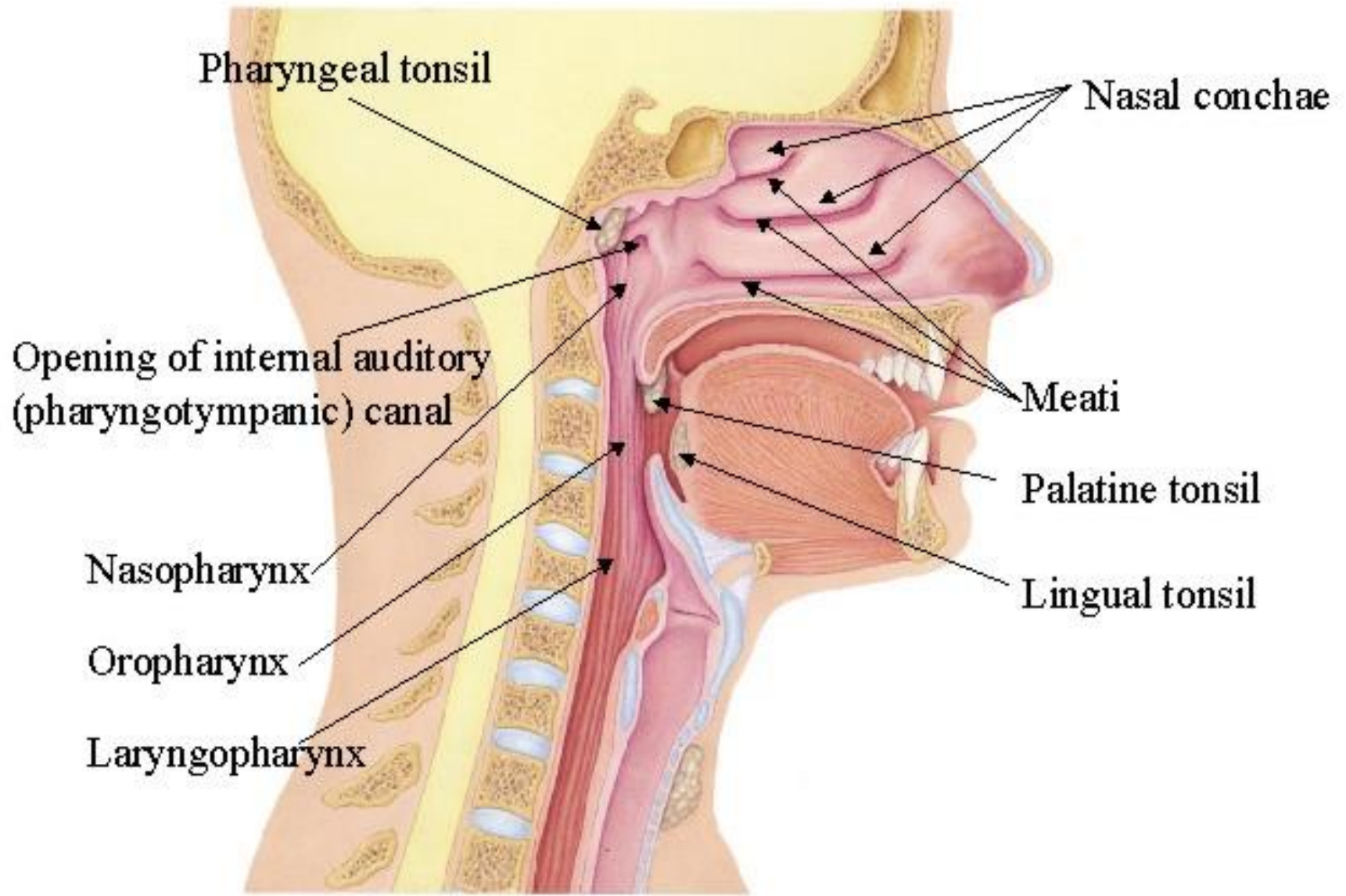
- Palatine tonsils are located in the lateral walls of the **oro-pharynx**.
- Lined with **a stratified squamous epithelium** that often densely infiltrated by lymphocytes.
- The lymphoid tissue contains **free lymphocytes** and **lymphoid nodules, generally with germinal centers**.
- Each tonsil has a number of **crypts**, which contain desquamated epithelial cells, live and dead lymphocytes, and bacteria.
- Crypts may appear as purulent spots in tonsillitis.
- The lymphoid tissue is **incompletely enclosed with a dense connective tissue capsule** which acts as a barrier against spreading of infections.

Palatine Tonsils

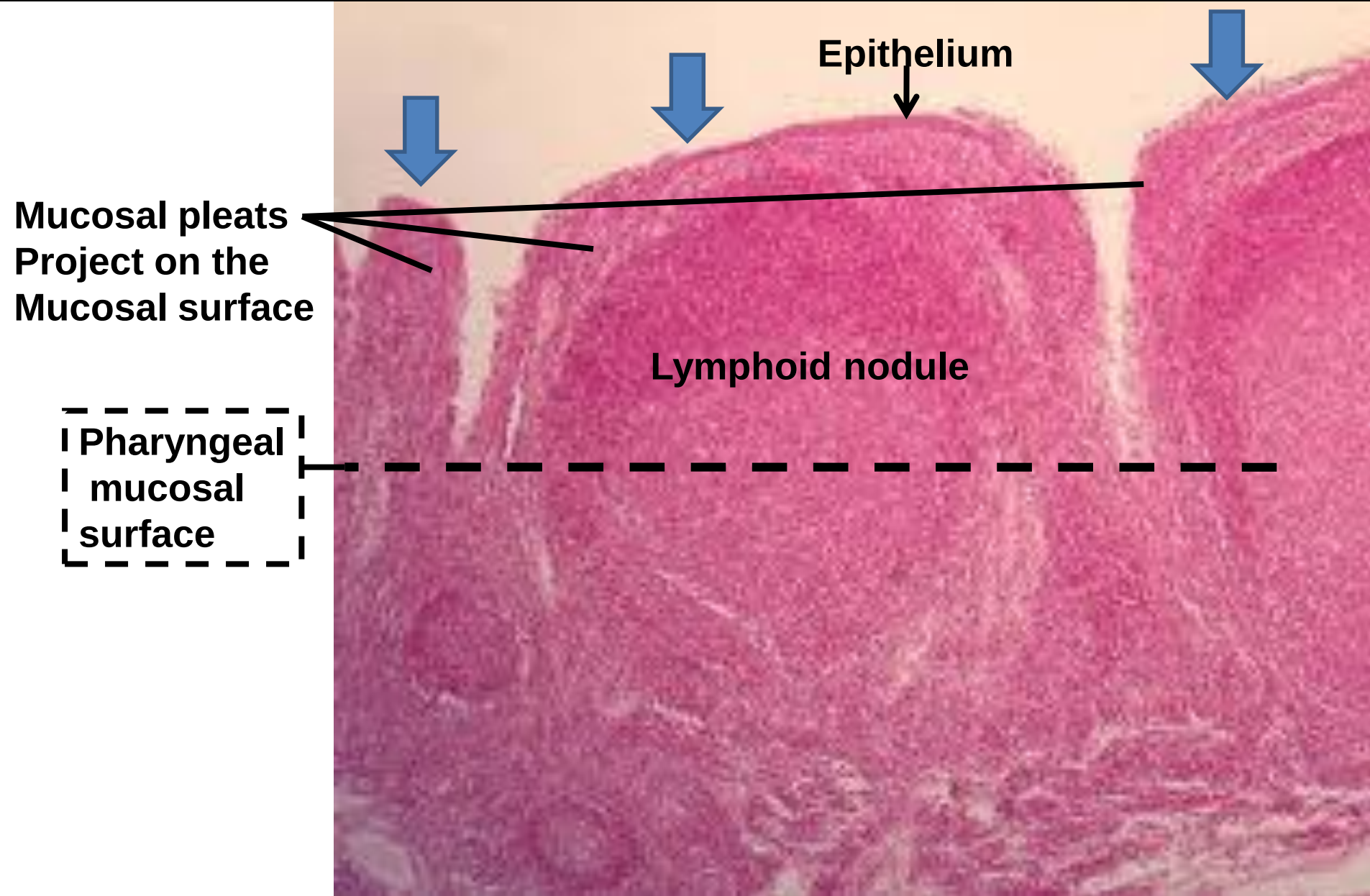
- (1) Crypt
- (2) stratified squamous epithelium
- (3) lymphoid nodules
- (4) diffuse lymphoid tissue
- (5) germinal center.
- (6) Capsule.
- (7) mucous glands.



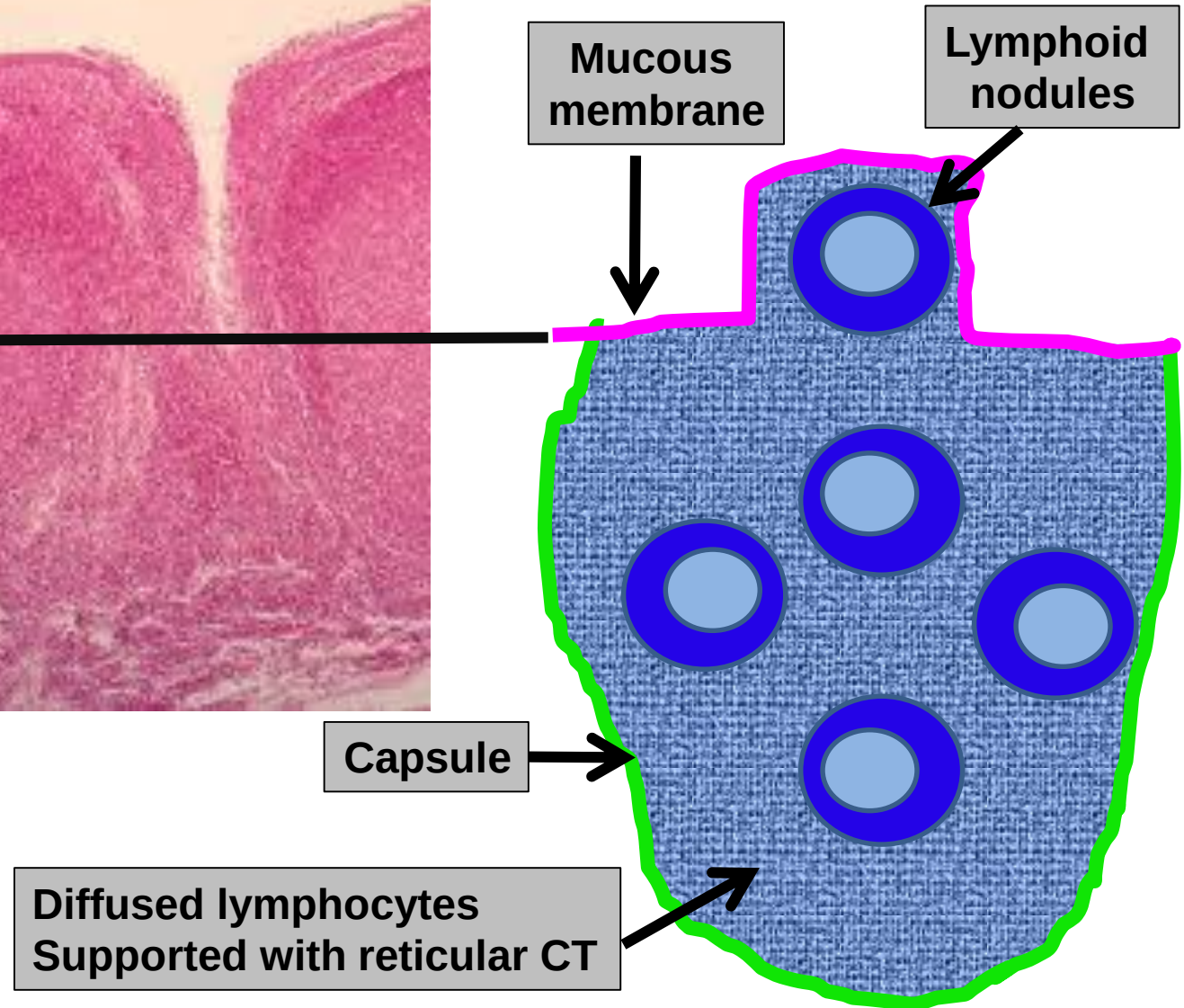
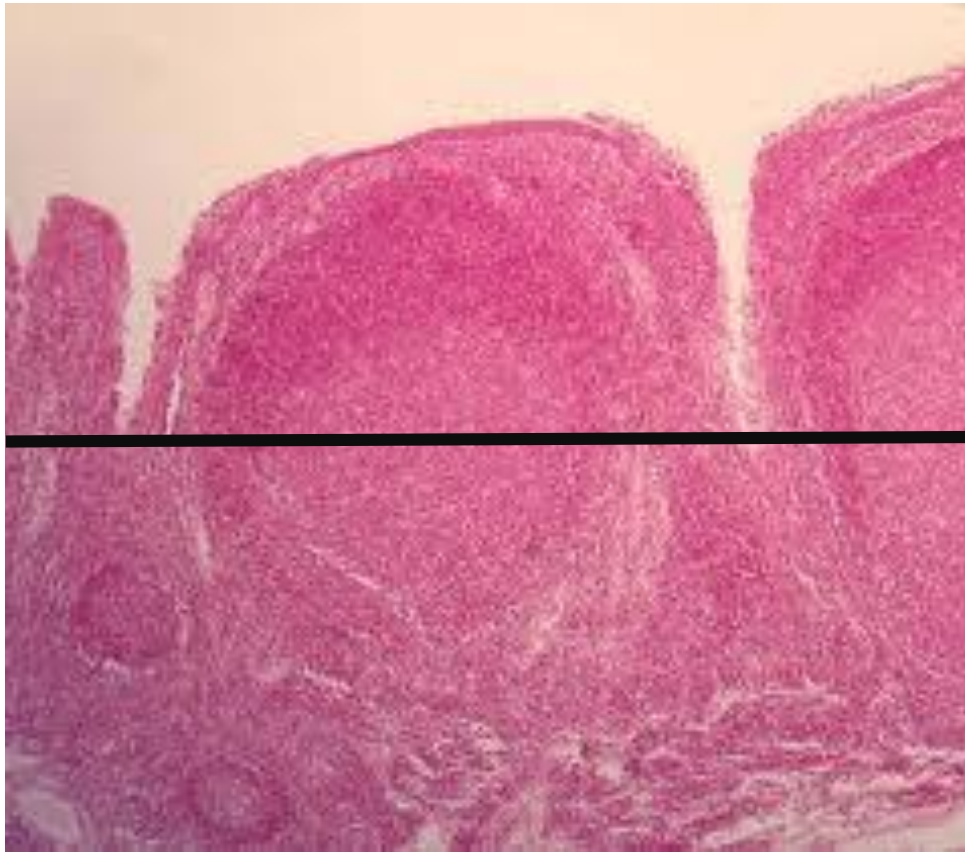
Pharyngeal Tonsil



Pharyngeal Tonsil



Pharyngeal Tonsil



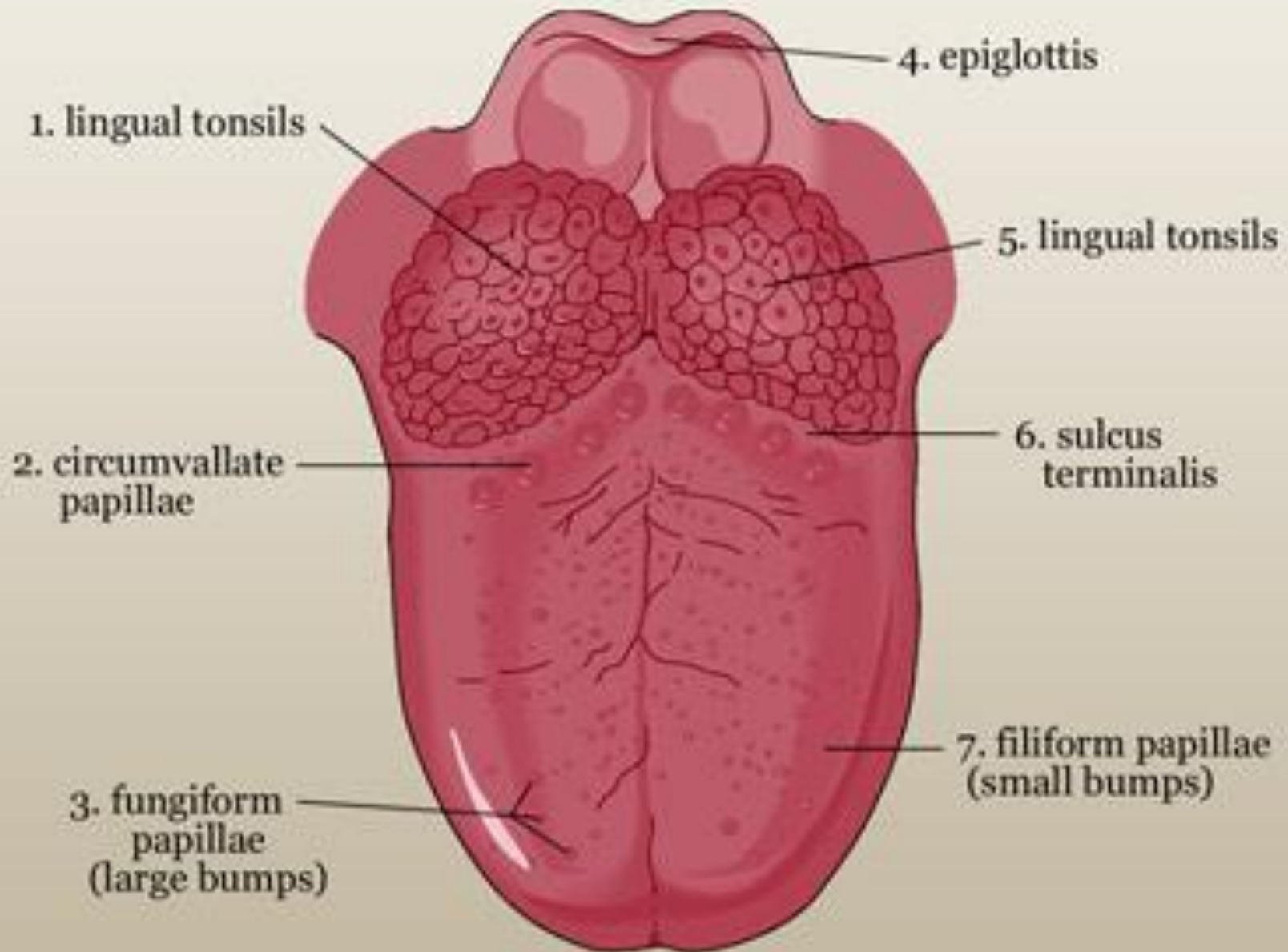
Pharyngeal Tonsil

- **Single tonsil** situated in the **dorsal- caudal portion of the pharynx**.
- Covered by **pseudostratified columnar ciliated epithelium** (typical of the respiratory tract).
- Composed of **pleats of mucosa**.
- **Has no crypts**
- Contains diffuse lymphoid tissue and lymphoid nodules.
- Its capsule is thinner than the capsule of the palatine tonsils.
- Hypertrophied pharyngeal tonsils resulting from chronic inflammation are called **adenoids**.

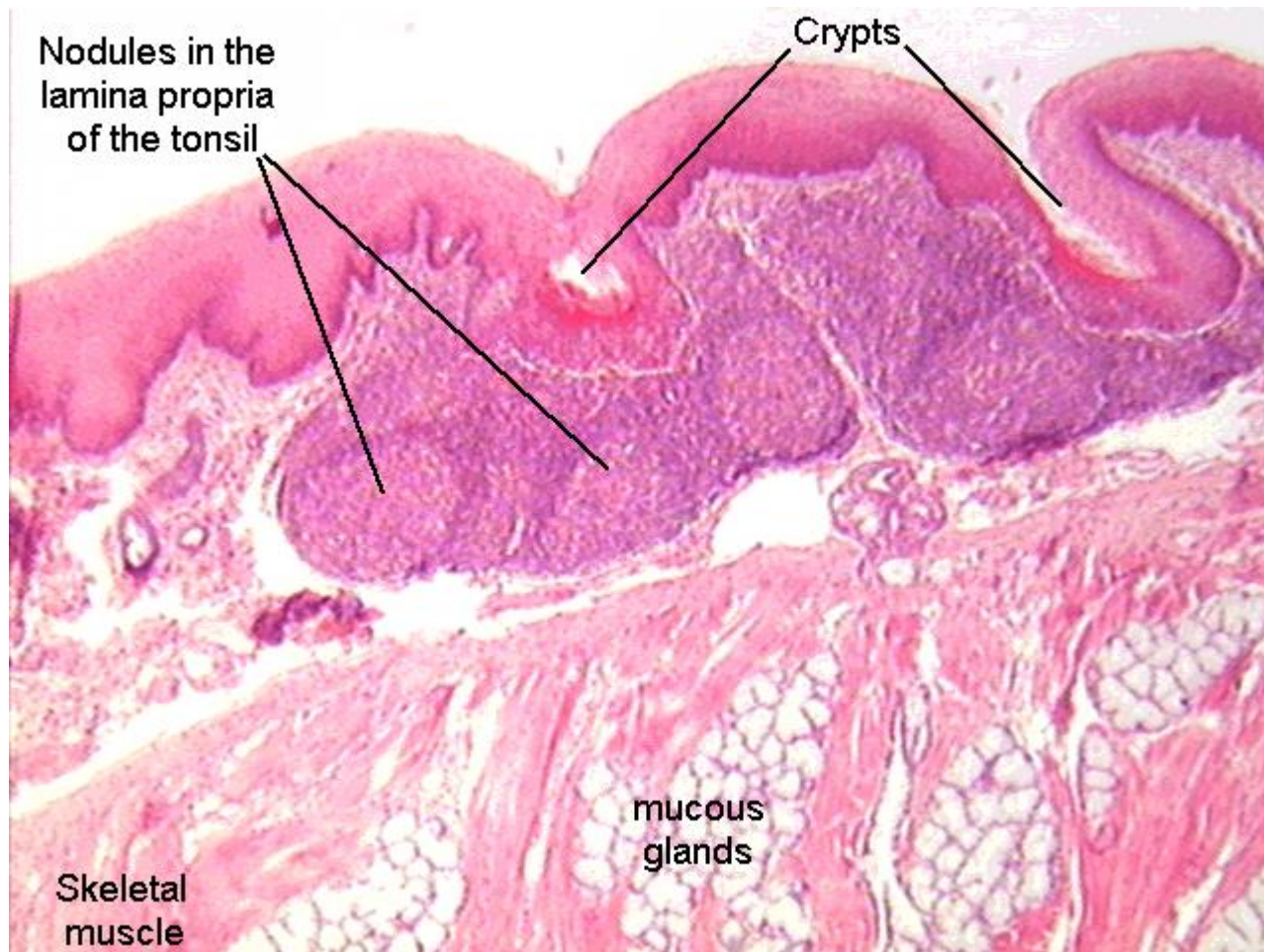
Lingual Tonsils

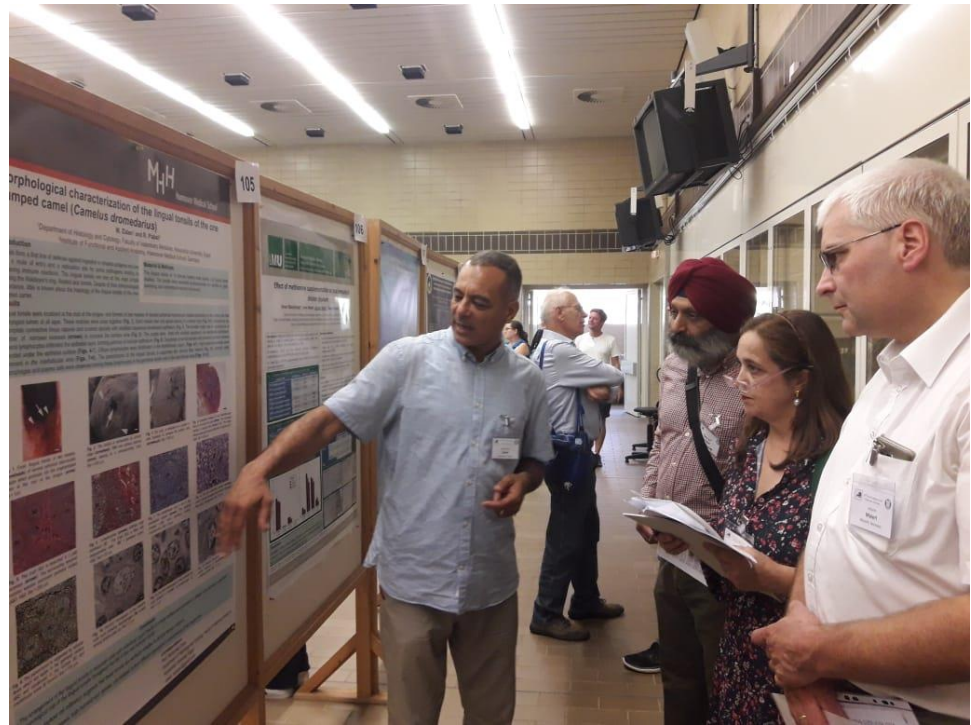
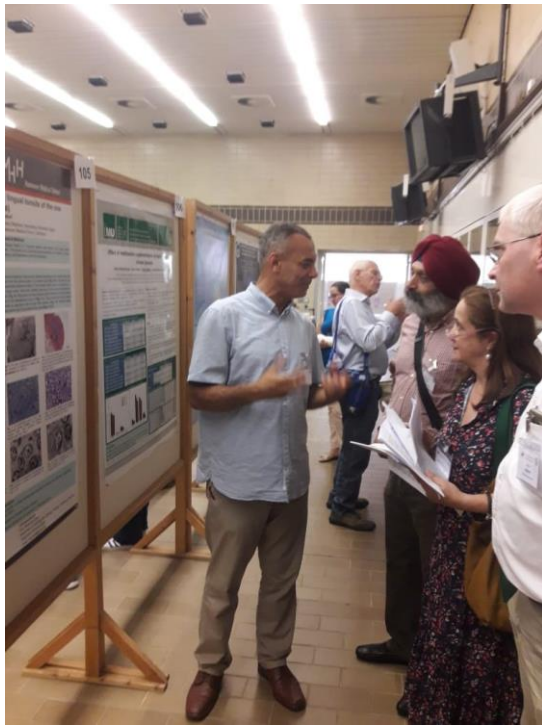
- **Smaller and more numerous** than the palatine and pharyngeal tonsils.
- Situated at the **base of the tongue**.
- Covered by **stratified squamous epithelium**.
- Each lingual tonsil has a single crypt.

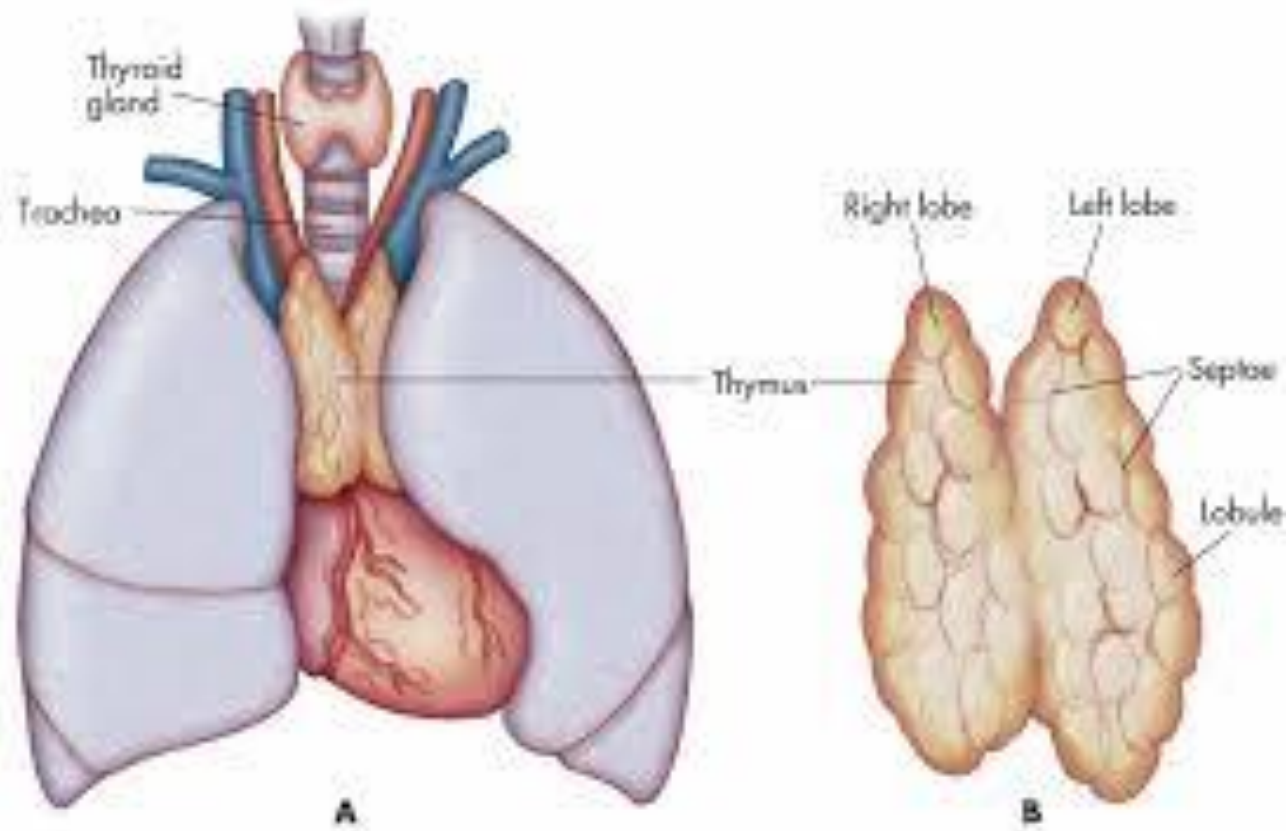
Lingual Tonsils



Lingual Tonsils



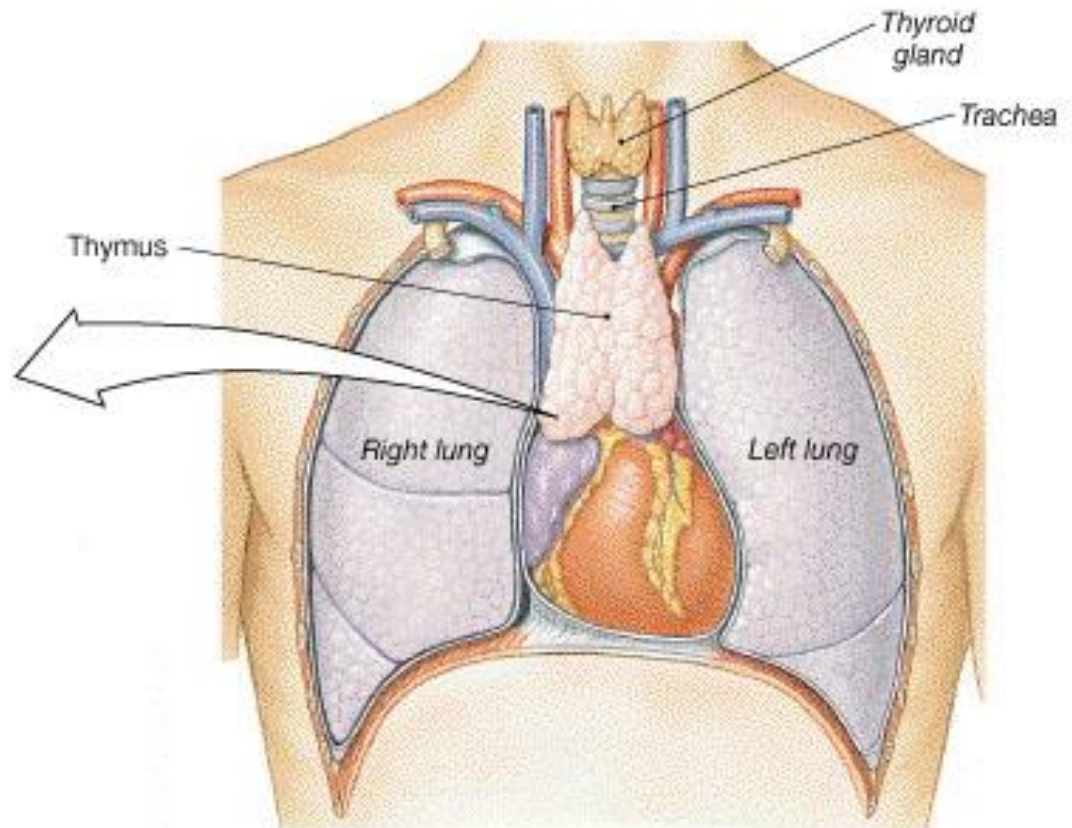
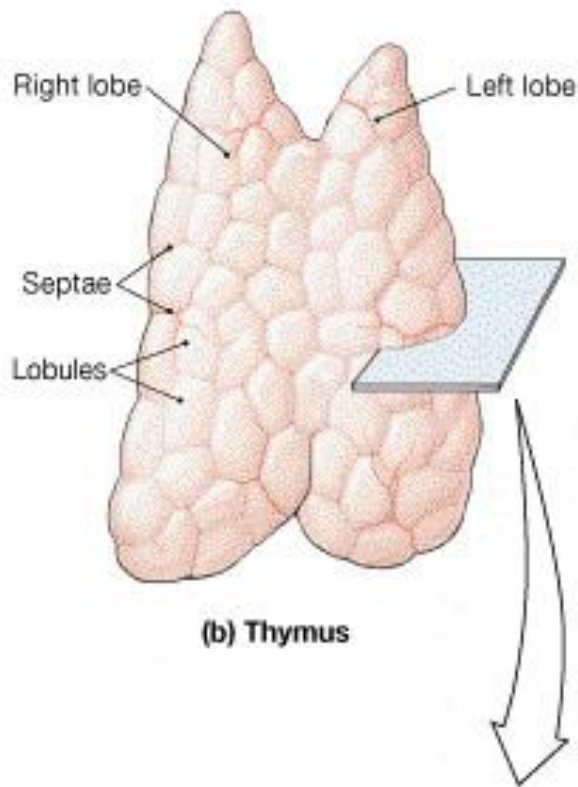




Thymus

Thymus

Bilobed structure lying in the thorax between the sternum and the pericardium.



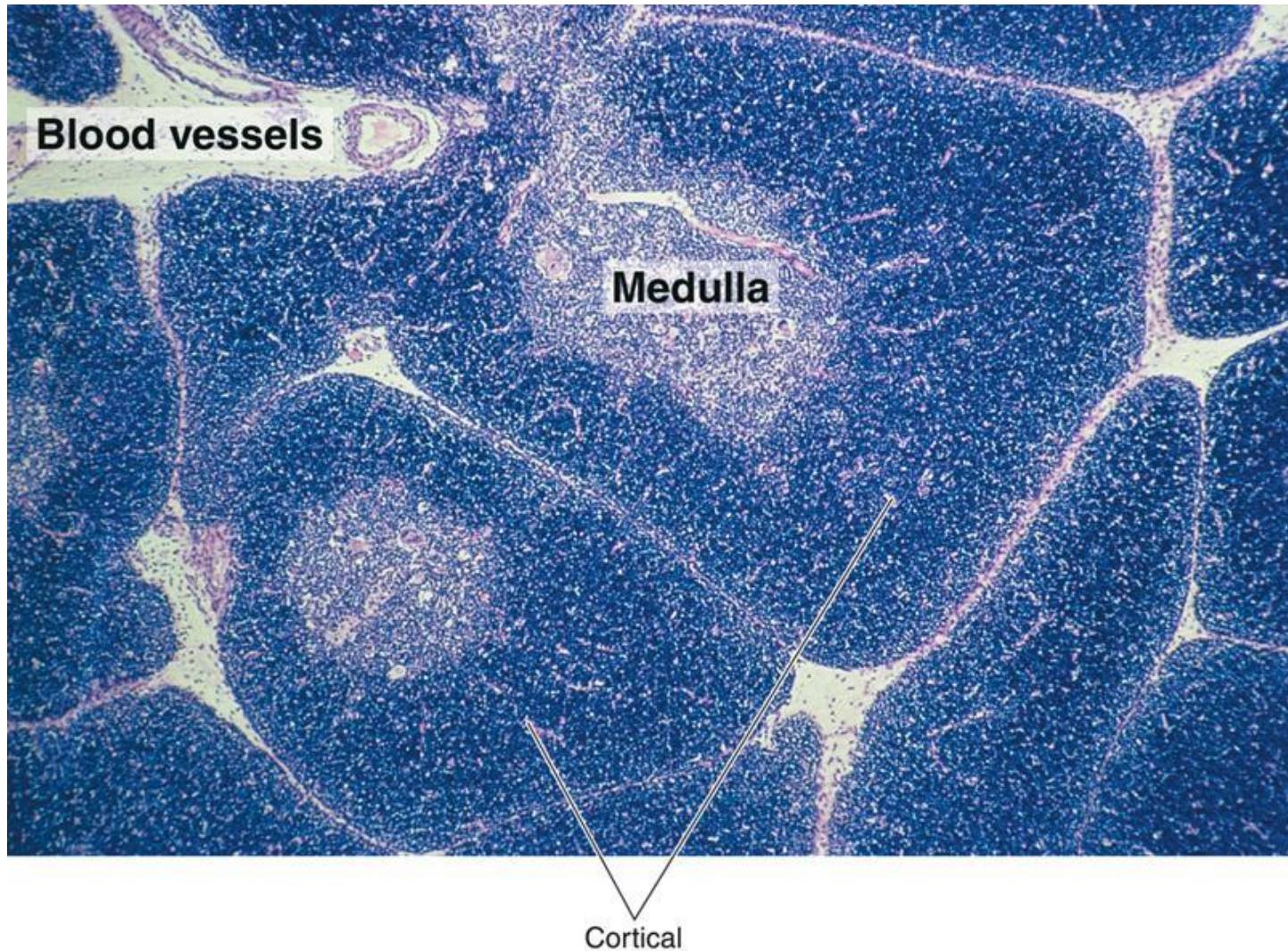
(a) Location of thymus within thoracic cavity

Thymus

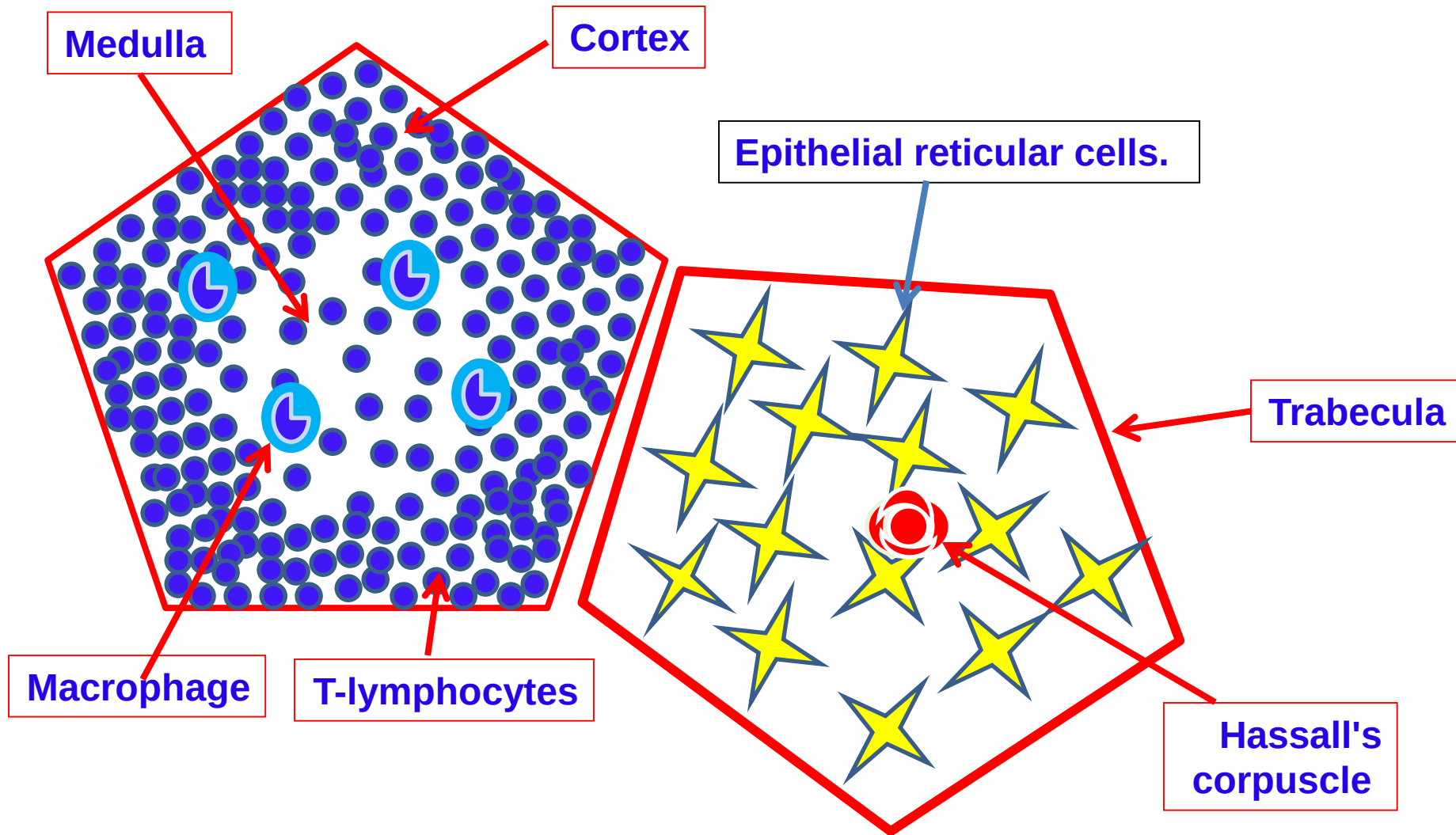
- In the newborn, it is at its greatest size relative to the size of the body. **It continues to grow until puberty** after that it **undergoes slow involution.**
- Surrounded by a delicate connective tissue capsule & trabeculae partially subdividing each lobe into lobules.
- **The thymus has no afferent lymphatic vessels and does not constitute a filter for the lymph.**
- ***The few lymphatic vessels in the thymus are all efferent.***

Thymus

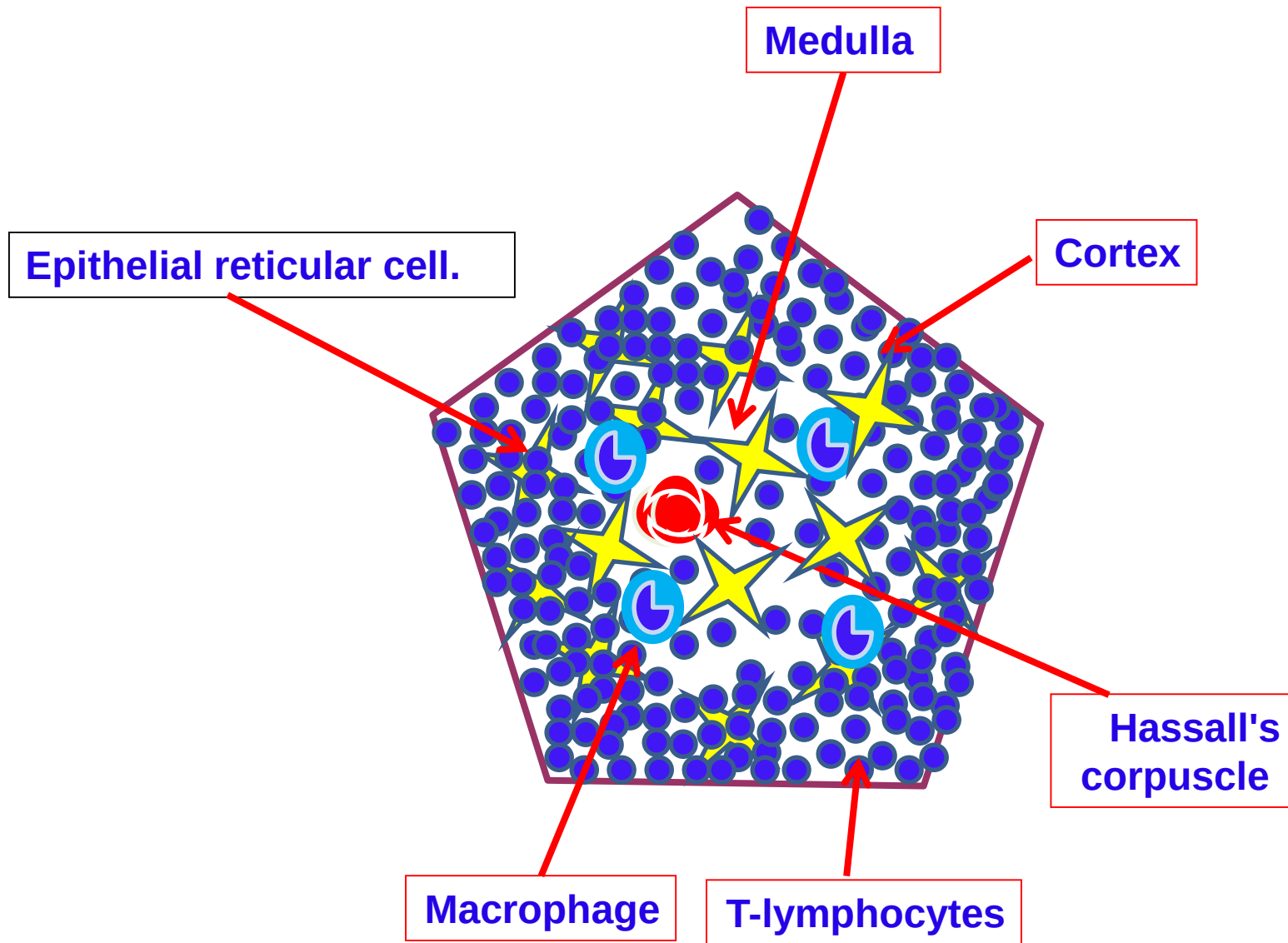
- Each lobule is composed of a **peripheral cortex** and **central medulla** and has a **framework of epithelial reticular cells**.



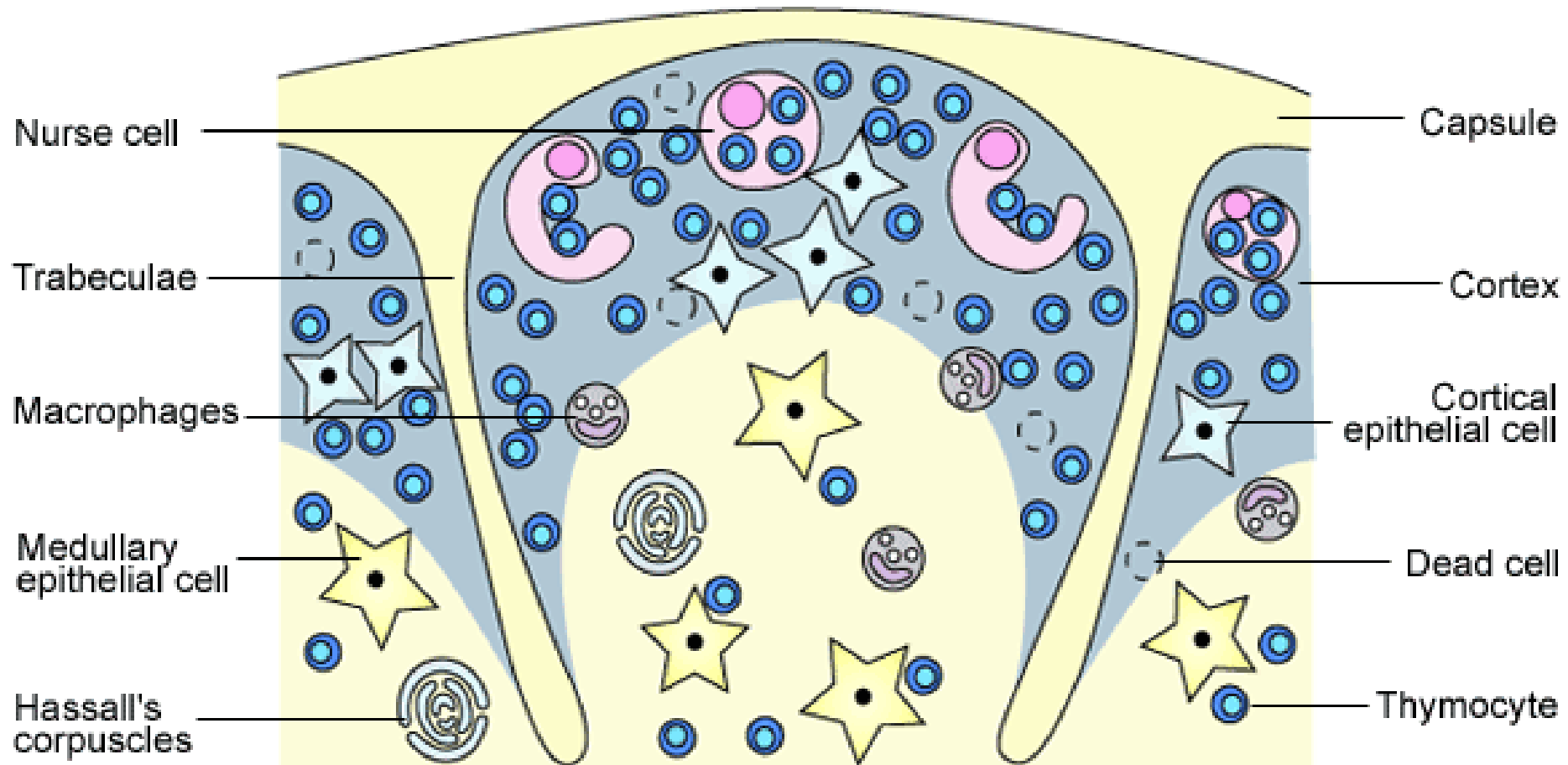
Thymic lobule



Thymic lobule



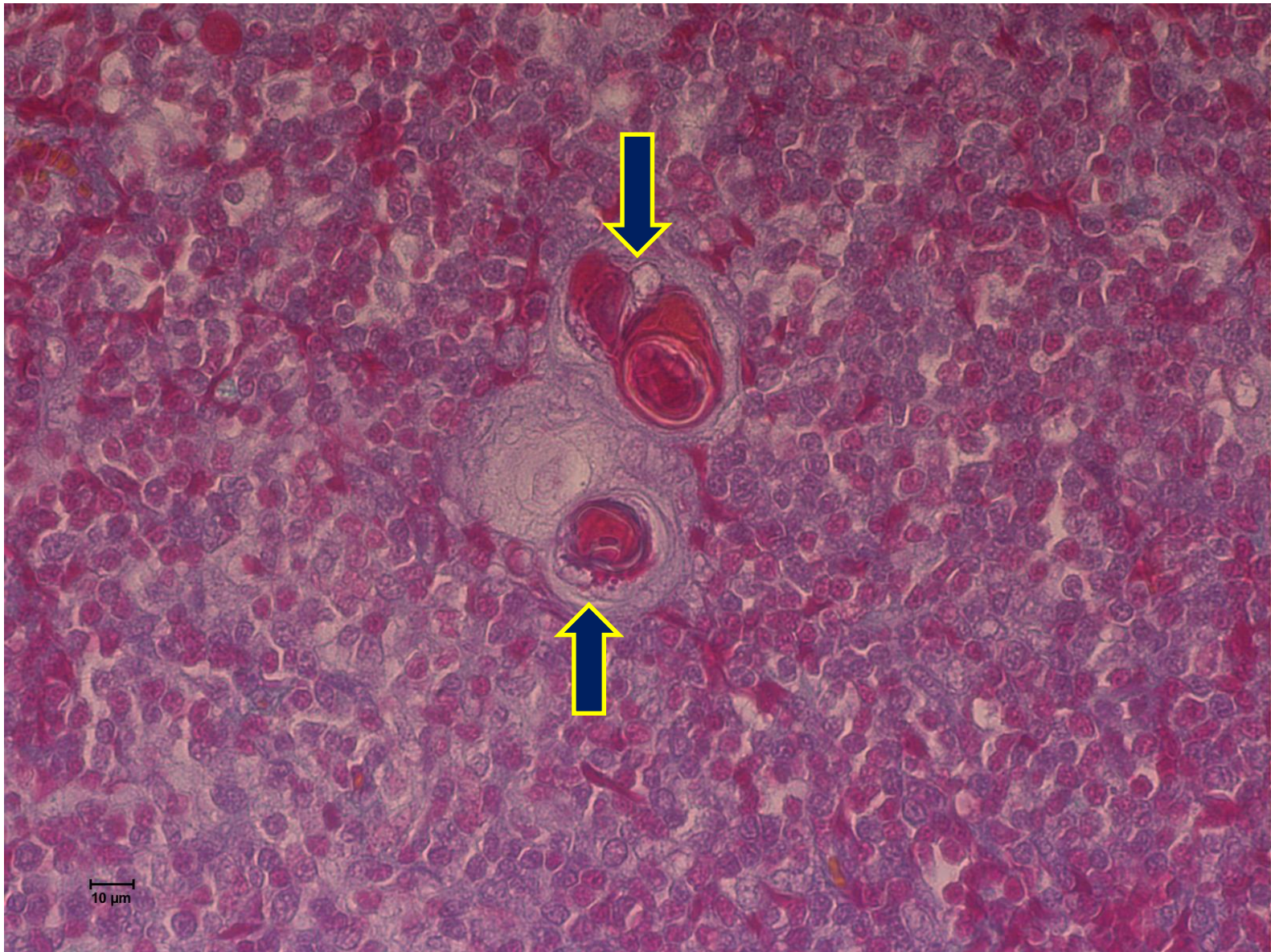
Thymic lobule



Thymus

- **In the cortex**, the lymphocytes are small and densely packed.
- **In the medulla** the lymphocytes are fewer.
- A characteristic structure known as **Hassall's corpuscles** are present in the medulla.
- **Hassall's corpuscles** are rounded structures stained red with eosin and consists of **flattened reticular epithelial cells concentrically wrapped around each other**.
- **The Hassall's corpuscle** represents degenerated reticular epithelial cells.

Hassall's corpuscles



Thymus

- **Lymphocytes** are continuously produced in the **cortex** by the division of lymphocytes or lymphoblasts.
- Many of these cells **die and phagocytosed** by **macrophages** in the thymus.
- **The reminders migrate toward the medulla** and enter the blood stream through the capillary walls.

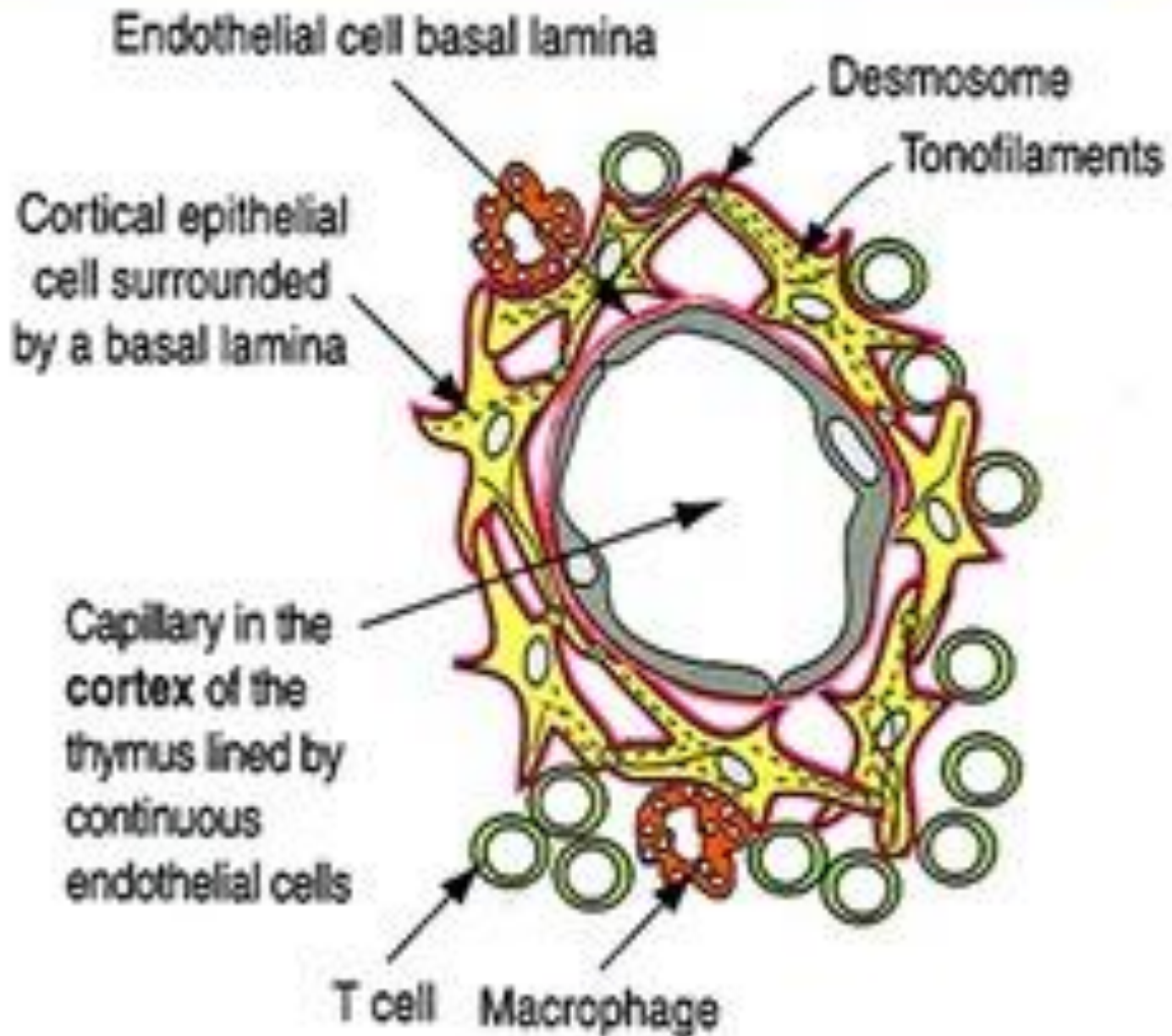
Thymus

- The lymphocytes of the thymus become functional when they moved into the blood or other lymphatic organs.
- The thymus lymphocytes (T lymphocytes) are responsible for cellular immunity.
- Arterioles and capillaries in the thymus are surrounded by processes of epithelial reticular cells.

Blood thymus barrier

- This barrier is located in **the cortex** to prevent most circulating antigens from reaching the thymus cortex and formed as follow:
 - 1. Cortical capillaries have a nonfenestrated endothelium.**
 - 2. Cortical capillaries have a very thick basal lamina.**
 - 3. The reticular epithelial cells form a sheath around the cortical capillaries.**
- This barrier makes the blood vessels of the thymus cortex impermeable to circulating antigens.

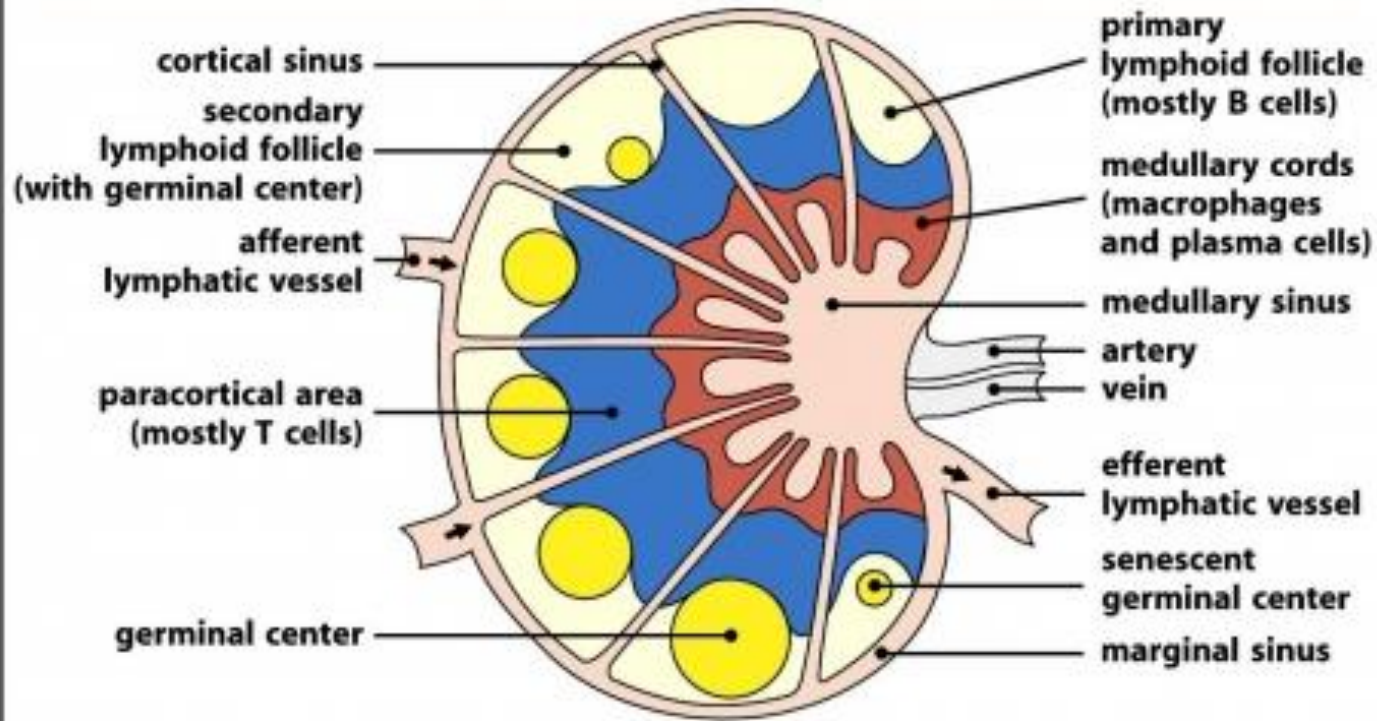
Blood thymus barrier



Function of the thymus

1. **The thymus produces T lymphocytes** which populates peripheral lymphatic organs, such as lymph nodes and spleen.
2. **The thymus produces several proteins** that act as growth factors to stimulate proliferation and differentiation of T lymphocytes. (paracrine secretions), including: **thymosin- α** , **thymopoietin**, **thymulin**, and **thymus humoral factor**.

A lymph node



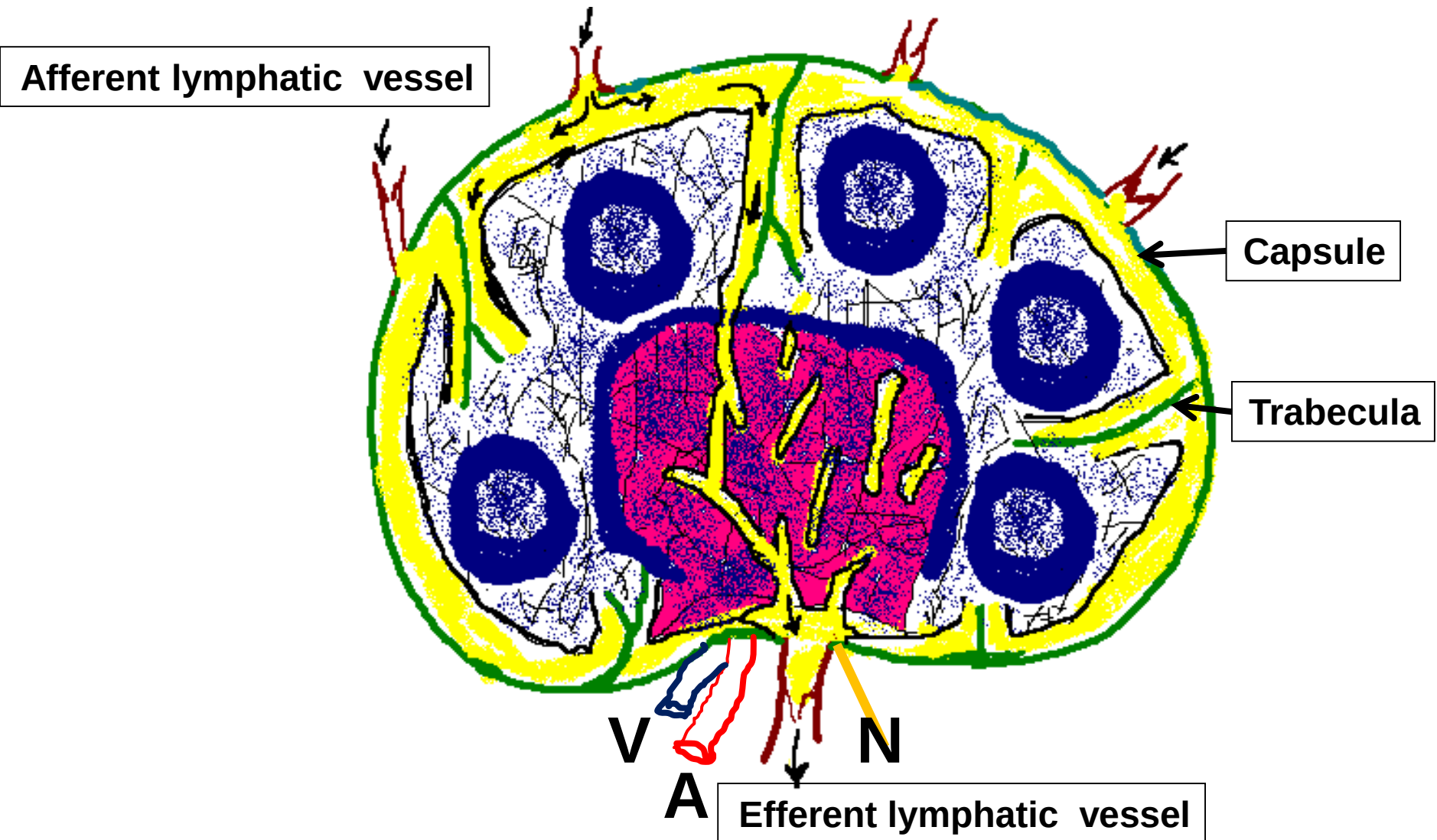
Lymph Nodes

Lymph Nodes

- **Constitute a series of filters against microorganisms and the spread of tumor cells.**
- **All lymph, derived from tissue fluid, is filtered by at least one lymph node before returning to the circulation.**

Lymph node

Lymph nodes are **elongated or kidney-shaped** organs.

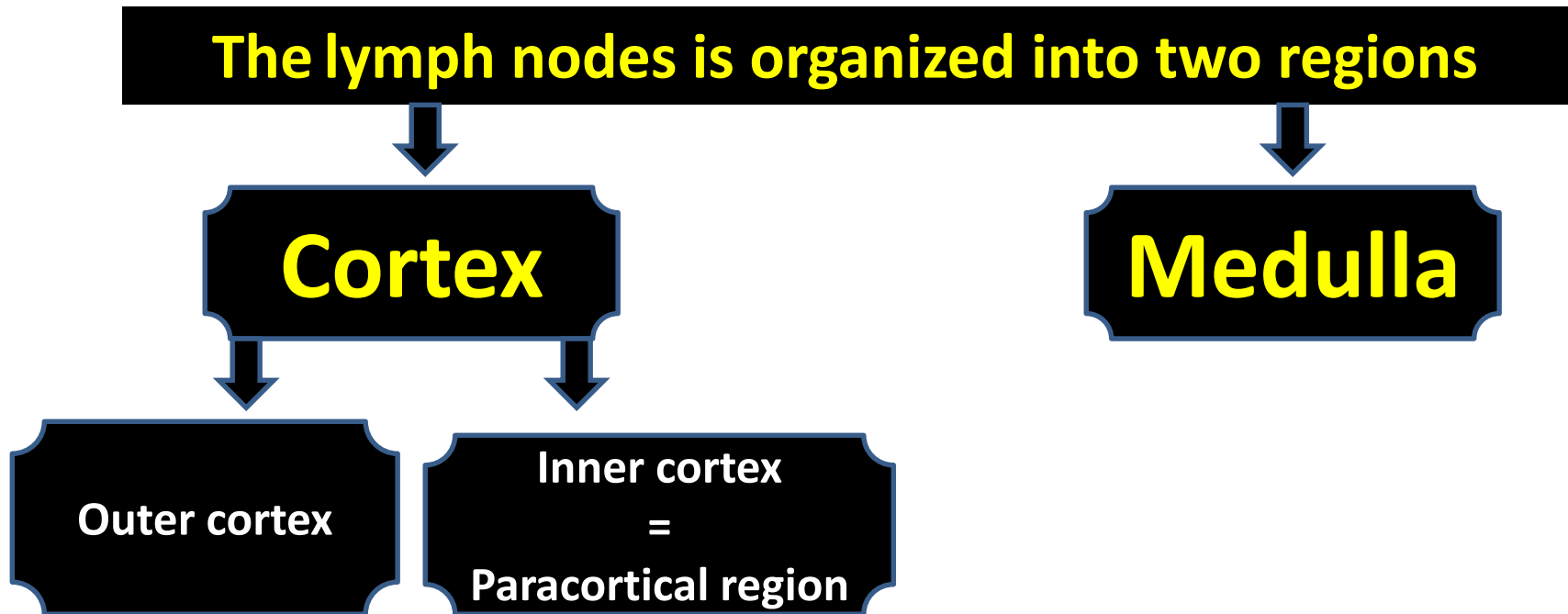


Lymph Nodes

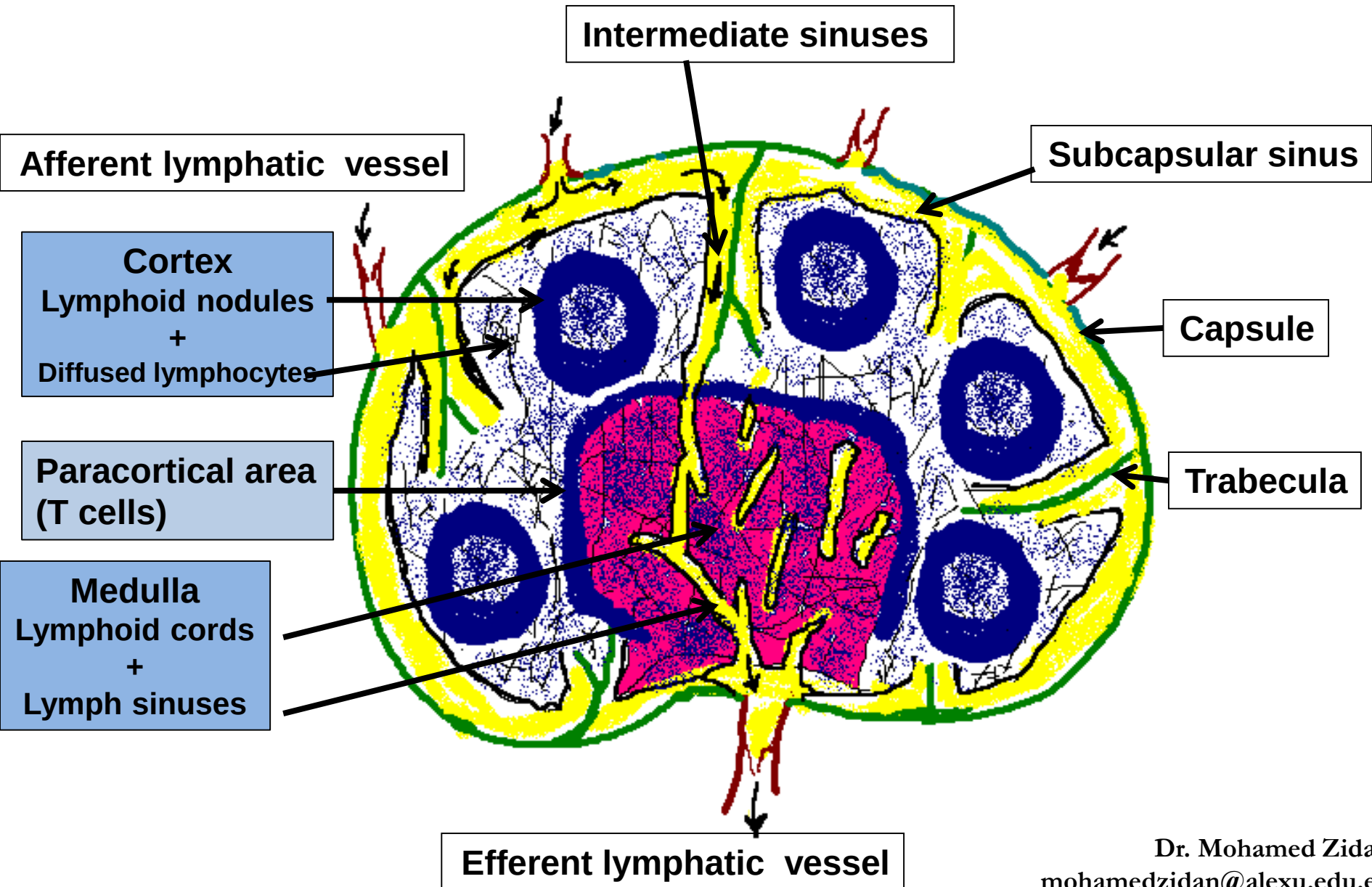
- Lymph nodes are **elongated or kidney-shaped** organs.
- Have a **convex** surface through which **lymphatic vessels** enter and a concave depression, **the hilum**, through which **arteries & nerves** enter, & **veins & lymphatic vessels** leave.
- **A connective tissue capsule** and trabeculae surround the lymph node.

Lymph Nodes

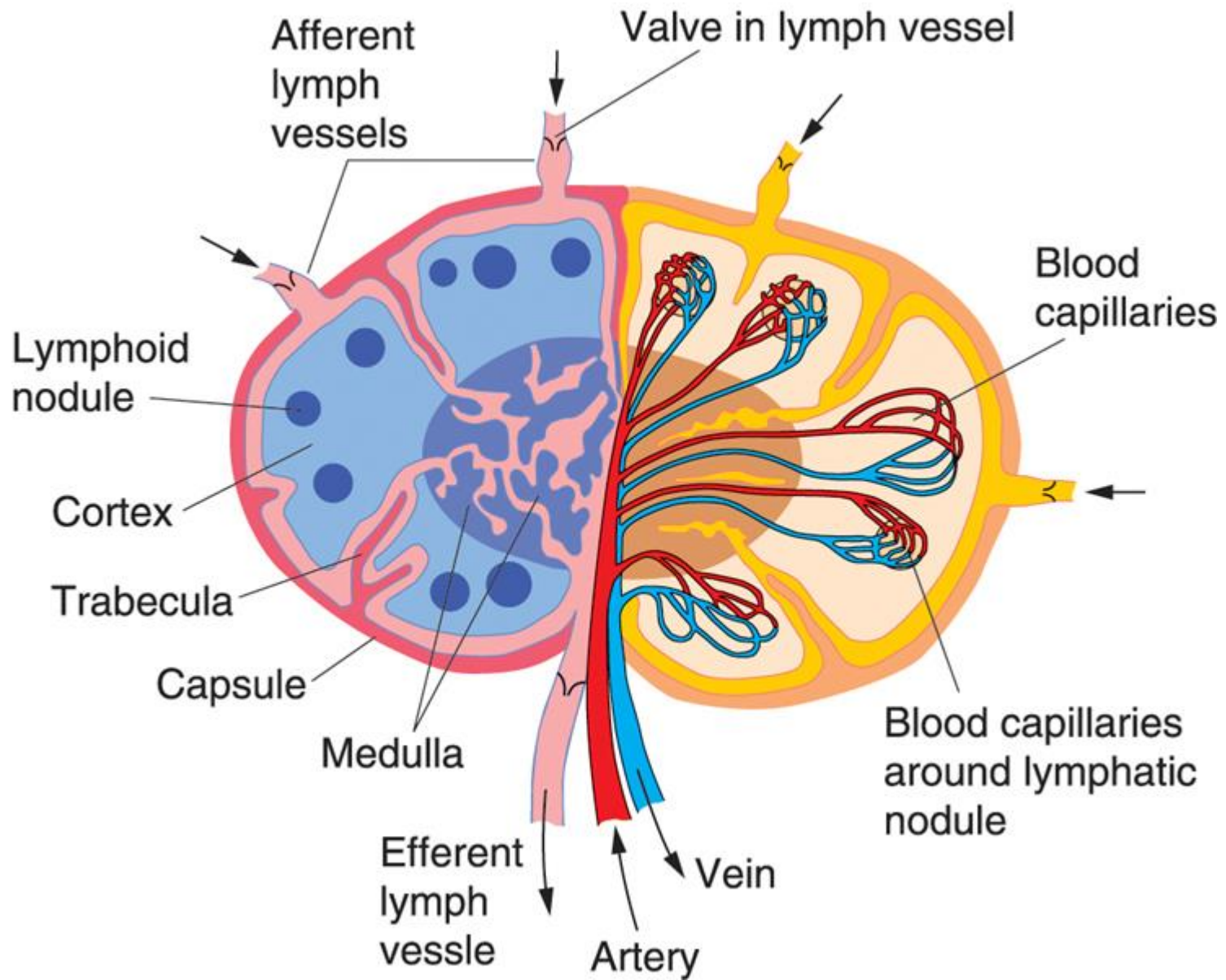
- The most common cells of lymph nodes are: **lymphocytes, macrophages and other APCs, plasma cells, reticular cells and follicular dendritic cells.**



Lymph node



Lymph Nodes



Lymph Nodes

The lymph nodes is organized into two regions

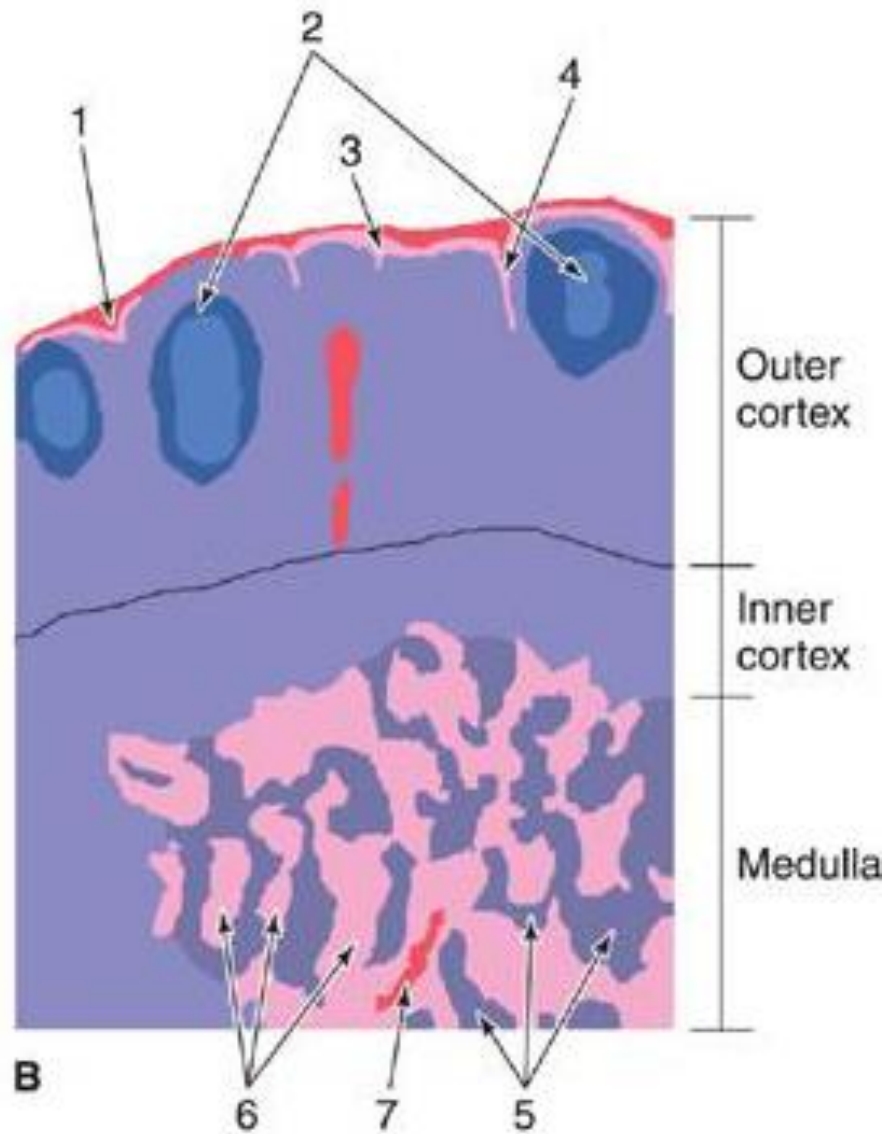
Cortex

Medulla

Outer cortex

Inner cortex
=
Paracortical region

Lymph Nodes



The outer cortex

Situated under the capsule, consists of:

- **1. A diffused population of cells** composed mainly of macrophages and APCs are also present. T lymphocytes and reticular cells .
- **2. Lymphoid nodules**, with or without germinal centers, formed mainly of **B lymphocytes**, embedded in the diffused population of cortical cells.

Cortex

- **3. Subcapsular sinuses** situated immediately beneath the capsule, They are composed of a loose network of reticular cells and fibers and lined with endothelium.
- *Lymph, containing antigens, lymphocytes, and APCs, circulates through these sinuses after being delivered into them by the afferent lymphatic vessels.*
- **4. Intermediate or radial sinuses** that run between lymphoid nodules. These sinuses arise from and have the same structure of the subcapsular sinuses.

The inner cortex (paracortical region)

- The inner cortex or paracortical region does not have precise boundaries with the outer cortex.
- Contains few, if any, nodules.
- Rich in T lymphocytes.

The inner cortex (paracortical region)

- Circulating T lymphocytes enter the lymph node in arterial blood then migrate through the wall of high endothelial venules into the paracortical zone; they rejoin the circulation after passing out of the lymph node through the efferent lymphatic vessel.

Medulla

The medulla



The medullary cords

The medullary sinuses

- **1. The medullary cords**
- Branched cordlike extensions of dense lymphoid tissue.
- Arise in the inner cortex.
- Contain **B lymphocytes** and often plasma cells and macrophages.

Medulla

2. The medullary sinuses

- Extended among the medullary cords.
- Are dilated spaces, frequently bridged by reticular cells and fibers.
- Contain lymph, lymphocytes, often many macrophages, and sometimes even granulocytes if the lymph node is draining an infected region.
- Arise from the intermediate sinuses and join at the hilum delivering the lymph to the efferent lymph vessel

Lymph Circulation

- **Afferent lymphatic vessels** cross the capsule and pour lymph into **the subcapsular sinus** to **intermediate sinuses** and, finally, into the **medullary sinuses**. During this passage, the lymph infiltrates the cortex and the medullary cords.
- The lymph is finally collected by **efferent lymphatic vessels** at the hilum.
- Valves in both the afferent and efferent vessels aid the unidirectional flow of lymph.

Lymph Circulation

Valves in the afferent and efferent vessels aid the unidirectional flow of lymph.

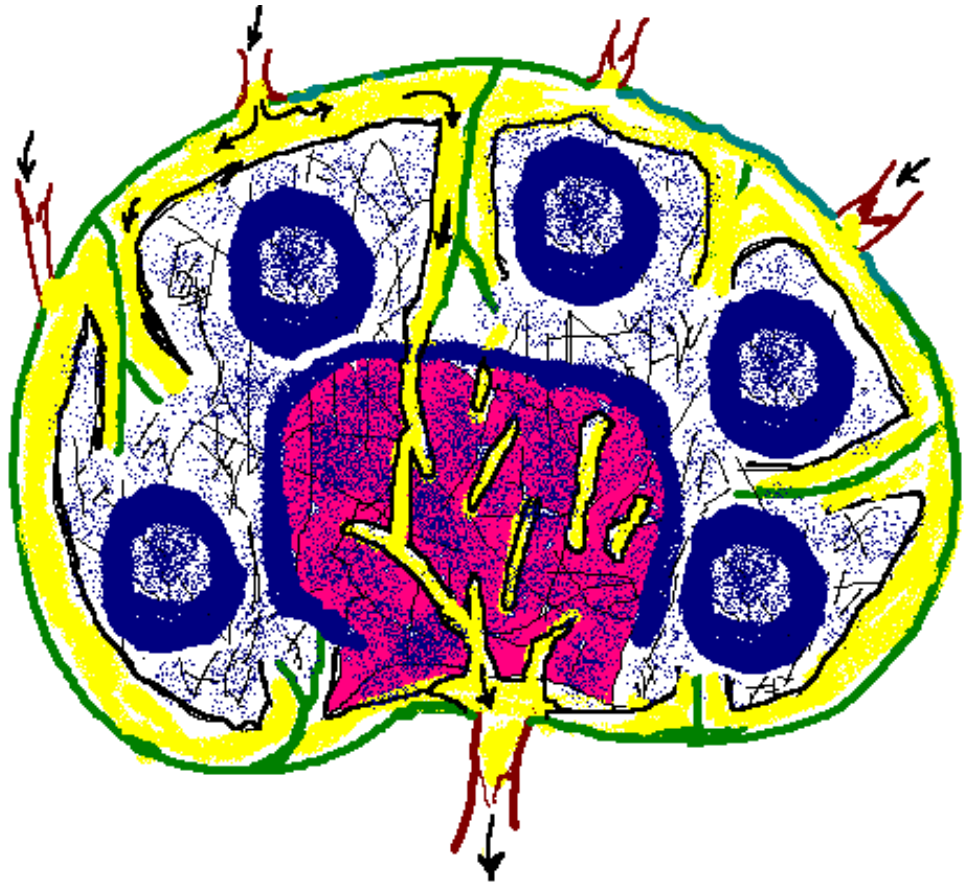
Afferent lymphatics

Subcapsular sinus

Intermediate sinuses

Medullary sinuses

Efferent lymphatics



Function of the Lymph Nodes

- 1. Filtration of the lymph.** Macrophages phagocytose foreign antigens.
- 2. Initiation of immune response.** The lymph that arrives at a lymph node may contain antigens, or antigens transported by APCs. All antigens have the opportunity to be presented to B and T lymphocytes to initiate an immune response.
- 3. Production of lymphocytes and plasma cells.** The lymph node is an important site of lymphocyte proliferation as well as of transformation of B lymphocytes into plasma cells. Because of this, the lymph that leaves a lymph node may be enriched in antibodies. As the lymph is transported to veins, these antibodies will be delivered to the entire body by the blood circulation.

Recirculation of Lymphocytes

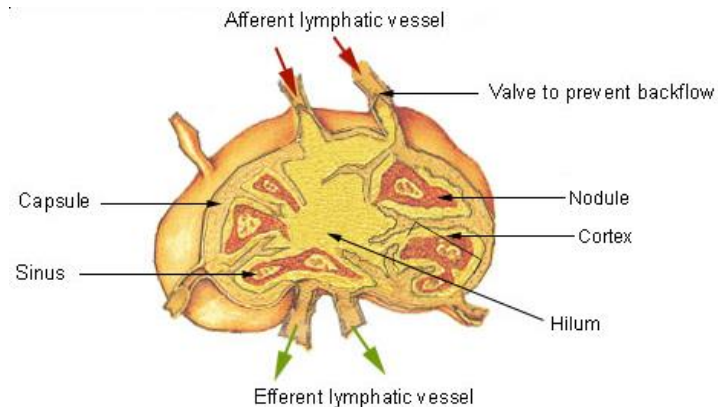
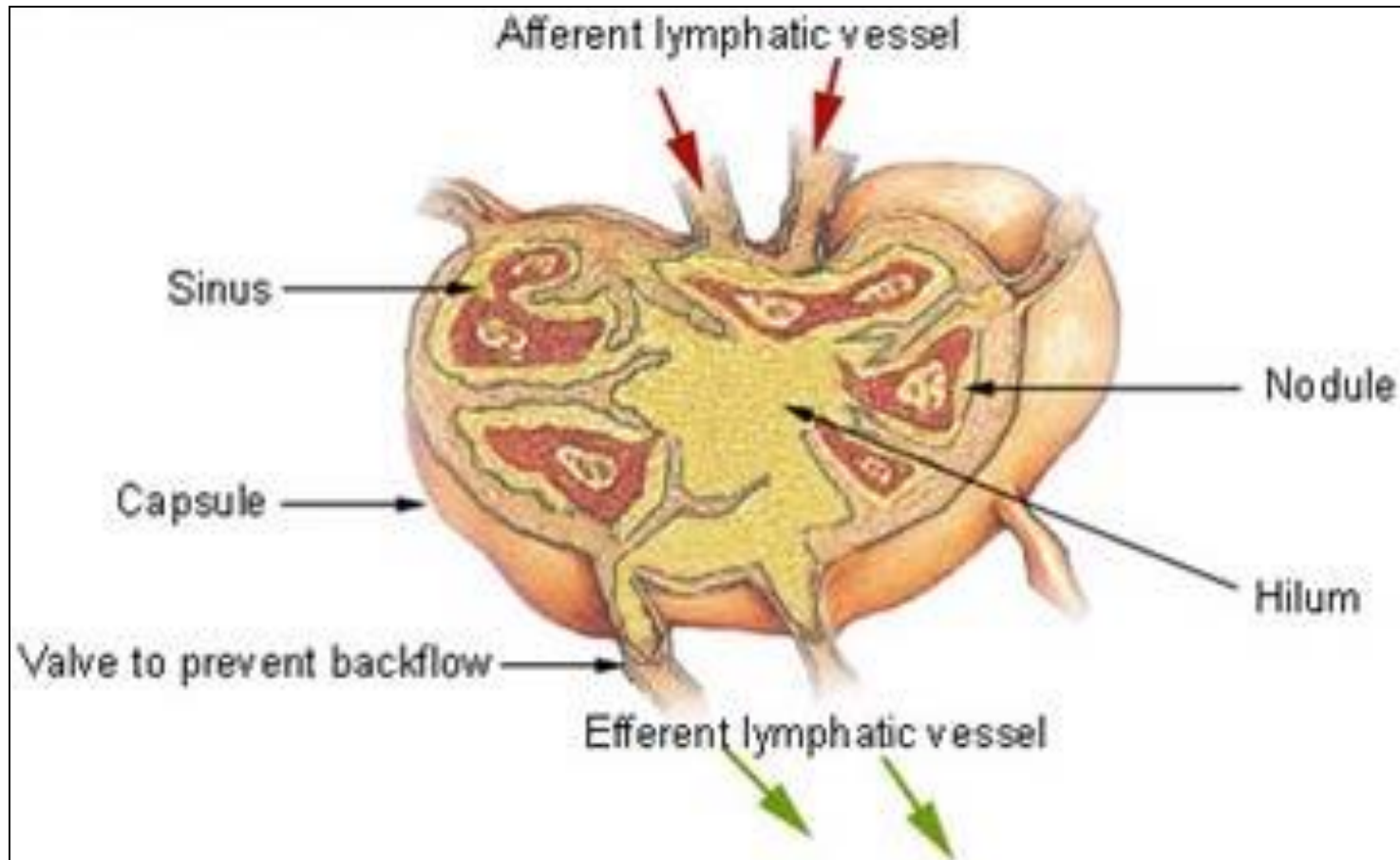
Lymphocytes that leave the lymph nodes by efferent lymphatic vessels eventually reach the bloodstream.

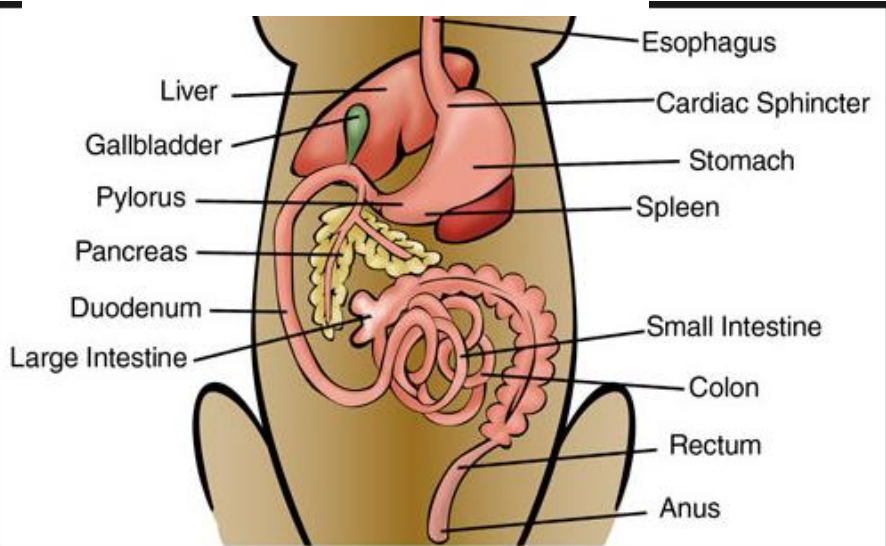
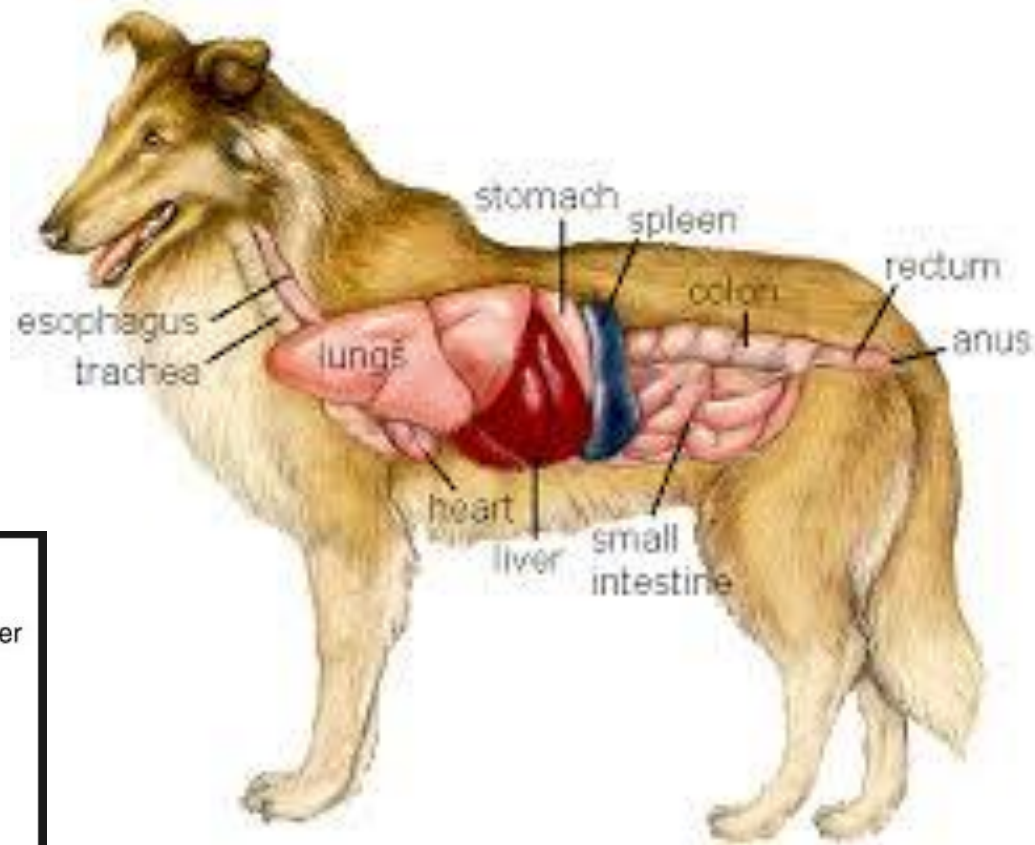
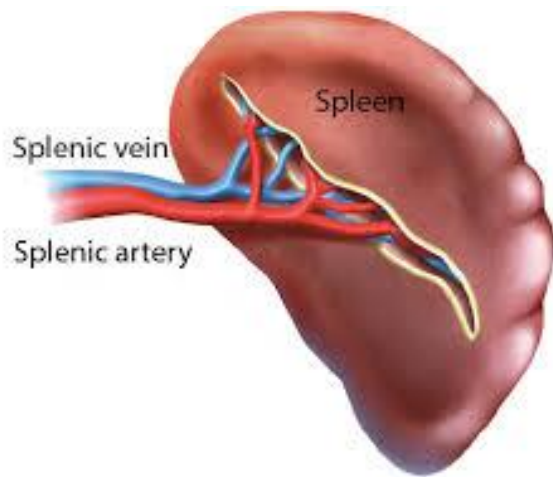
- They may then leave the blood vessels by entering the tissues and return to another lymph node by a lymph vessel.
- They may also return to a lymph node (a process called homing) by crossing the walls of specific blood vessels, the high endothelial venules (HEVs), present in lymph nodes.
- These venules have an unusual endothelial lining of tall cuboidal cells. High endothelial venules are also present in other lymphoid organs, such as the tonsils, and Peyer's patches, but not in the spleen.
- The continuous recirculation of lymphocytes enables most parts of the body to be constantly monitored, increasing the opportunity for lymphocytes to be in contact with APCs and antigens that have migrated to lymph nodes.

lymph nodes of pig

- The lymph nodes of pig are unique in its structure as it differs from all mammals in that the lymph nodules located deep in the medulla (not in the cortex) while the peripheral tissue (cortex) is unorganized (no lymphoid cords). One afferent vessel enters at the hilus and several efferent vessels leave the lymph node in the periphery.

lymph nodes of pig



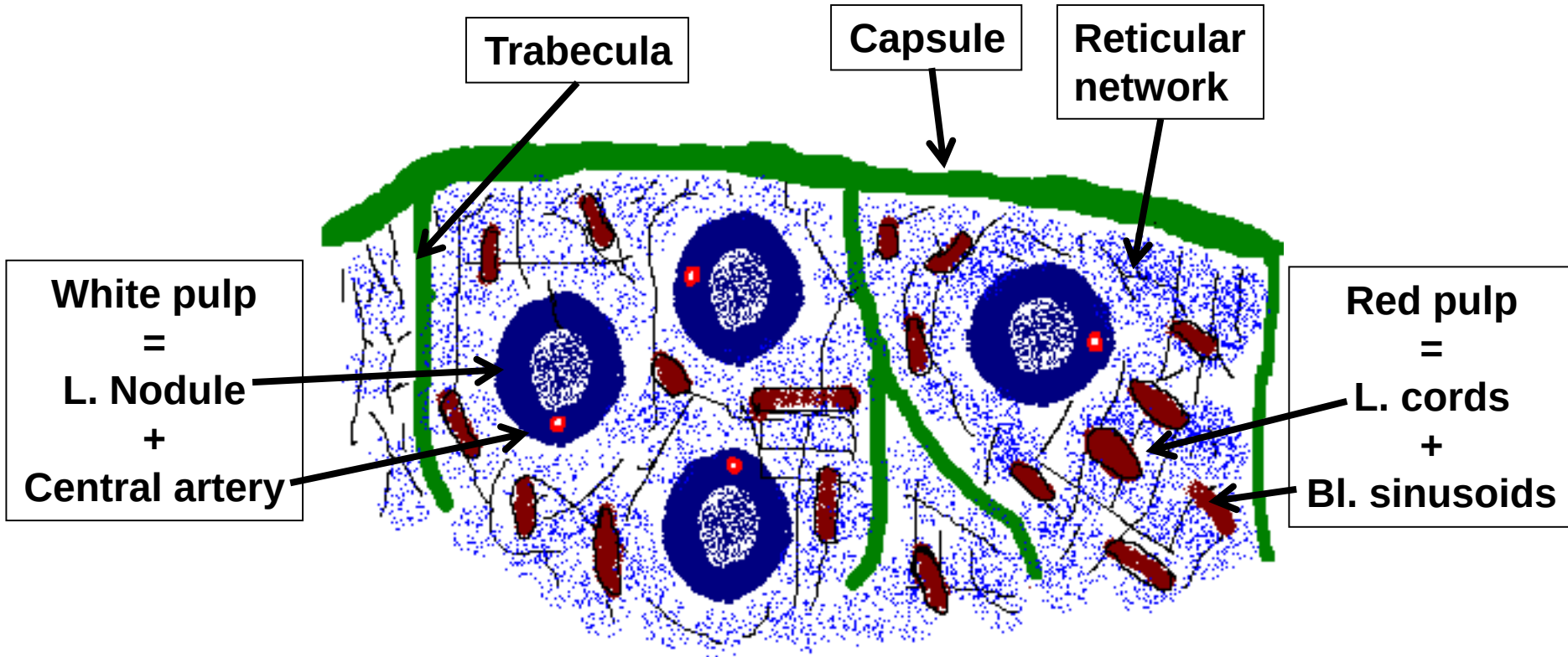


Spleen

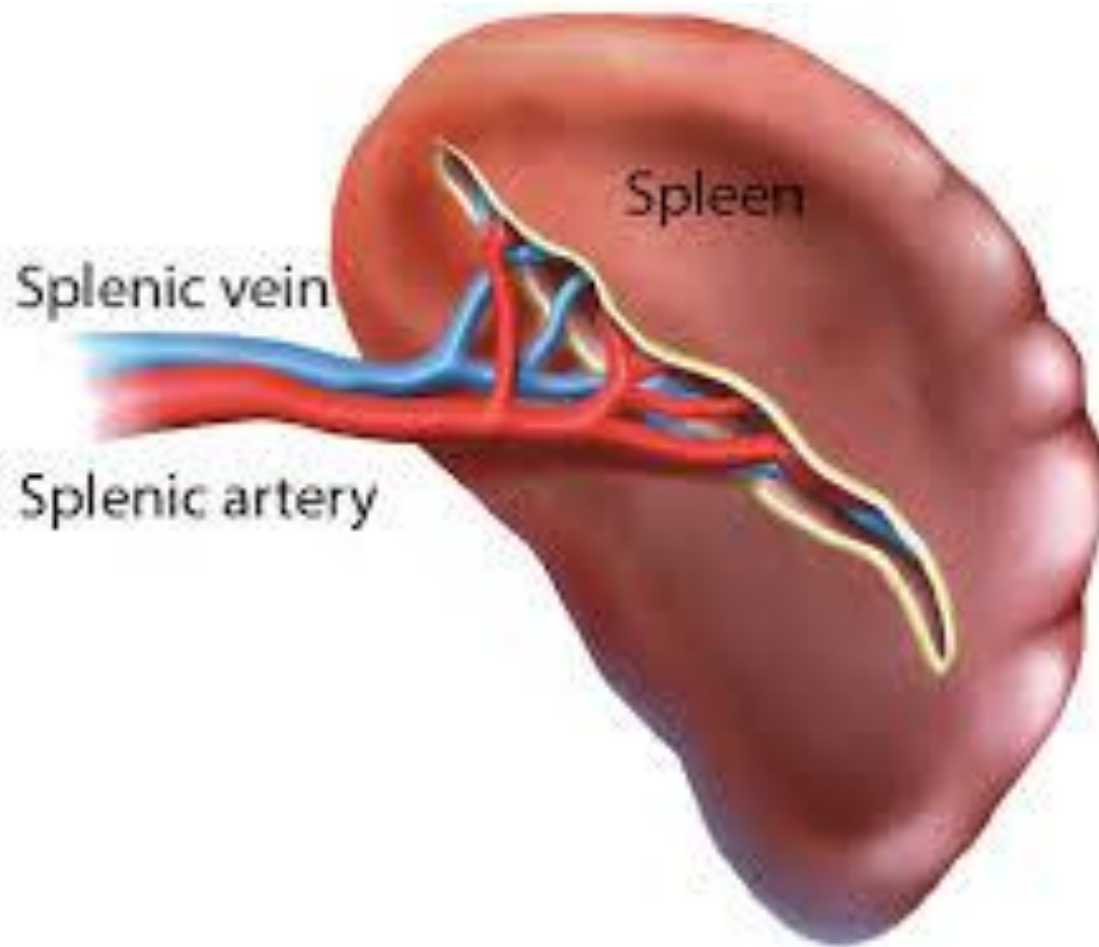
Spleen

- The spleen is **the largest lymphoid organ** in the body.
- It receives blood supply via the **splenic artery** and drained by the **splenic vein**.
- It is the site of destruction of aged erythrocytes.
- The spleen is a production site of activated lymphocytes, which are delivered to the blood.
- The spleen reacts directly to antigens carried in the blood and is, thus, an important blood filter and antibody-forming organ.

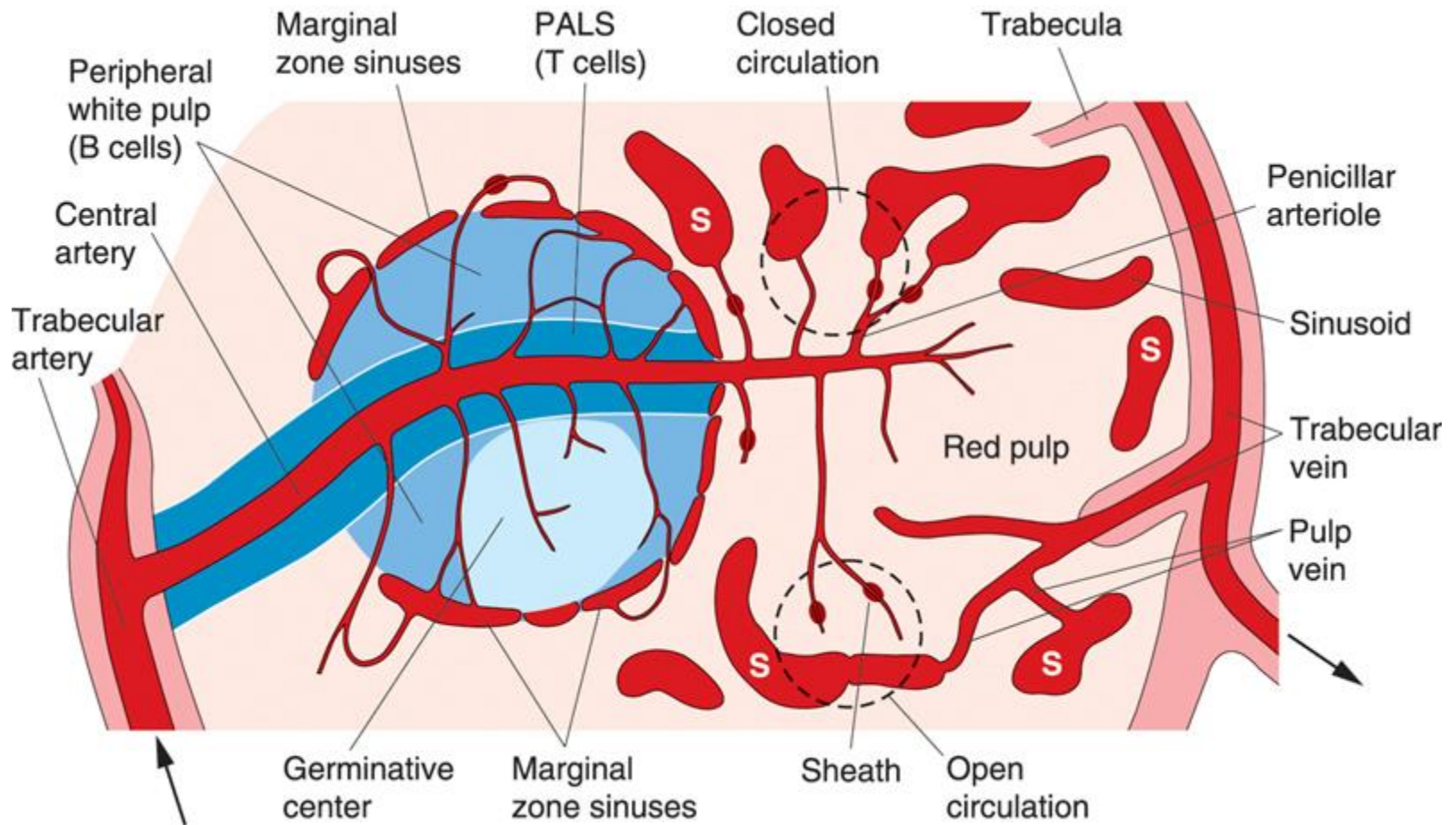
Spleen



Splenic Circulation

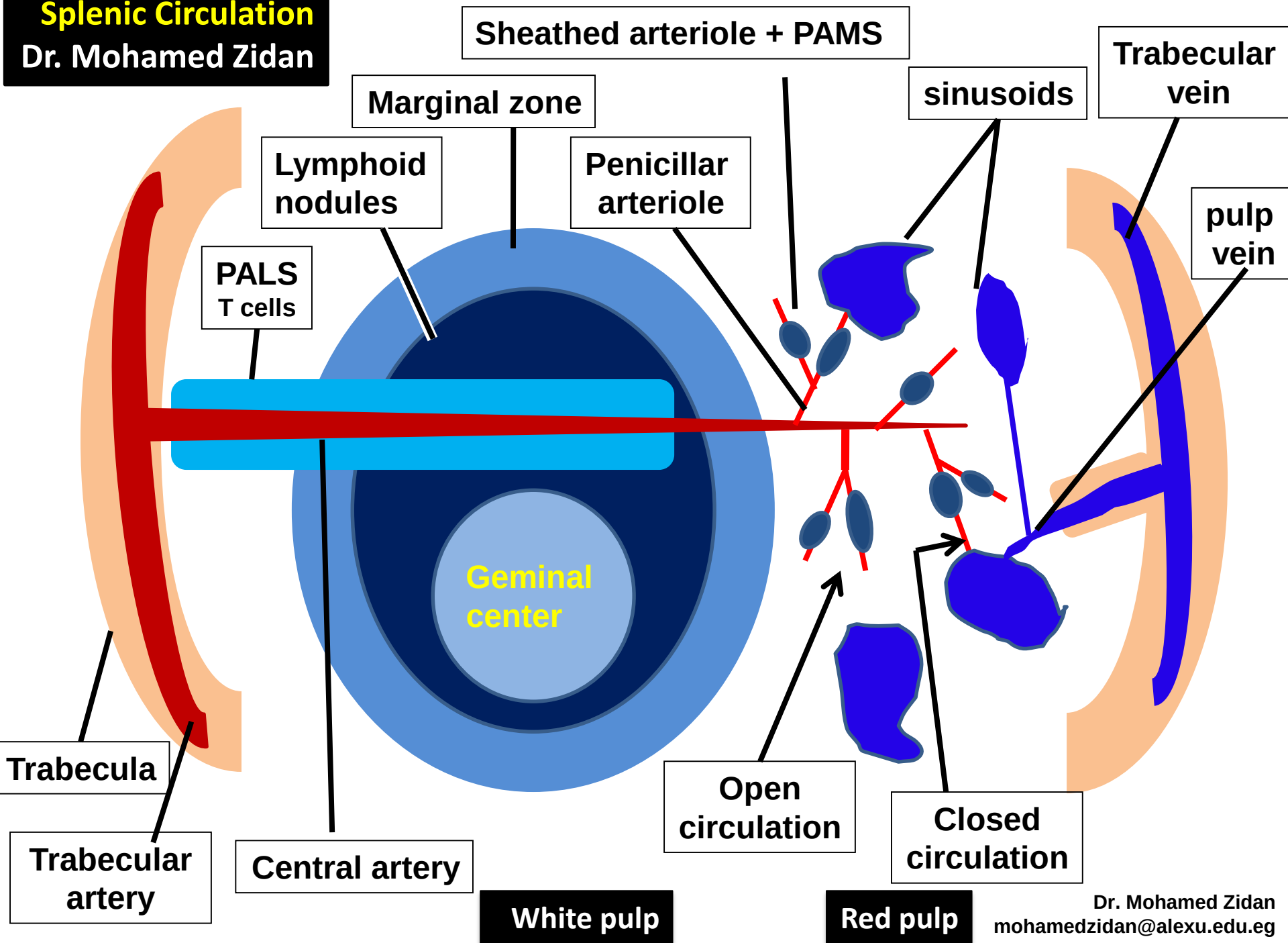


Splenic Circulation



Splenic Circulation

Dr. Mohamed Zidan



Sheathed arteriole + PAMS

Marginal zone

Lymphoid nodules

PALS
T cells

Penicillar arteriole

sinusoids

Trabecular vein

pulp vein

Geminal center

Trabecula

Trabecular artery

Central artery

Open circulation

Closed circulation

White pulp

Red pulp

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General Structure

- The spleen is surrounded by a **capsule** and **trabeculae** of dense connective tissue and smooth muscles, which divide the parenchyma, or **splenic pulp**, into incomplete compartments.
- Large trabeculae originate at the hilum, on the medial surface of the spleen; **these trabeculae carry nerves and arteries** into the splenic pulp as well as veins that bring blood back into the circulation. Lymphatic vessels that arise in the splenic pulp also leave through the hilum via the trabeculae.

General Structure

- The spleen is composed of a **network of reticular tissue that contains reticular cells, many lymphocytes and other blood cells, macrophages, and APCs.**
- The splenic pulp has two components, the **white pulp** and the **red pulp**.
 - **The white pulp** consists of the **periarterial lymphatic sheath (PALS)** and the **lymphoid nodules**,
 - **The red pulp** consists of **splenic cords (Billroth's cords)** and blood **sinusoids**.

White Pulp

- The splenic artery divides into **trabecular arteries** of various sizes that follow the course of the connective tissue trabeculae.
- When they leave the trabeculae to enter the parenchyma, the arteries are immediately enveloped by a sheath of T lymphocytes called the periarterial lymphatic sheath (**PALS**), which is part of the white pulp. These vessels are known as **central arteries** or **white pulp arteries**.
- After coursing through the parenchyma for variable stretches, the PALS receive large collections of lymphocytes (mostly B cells) forming lymphoid nodules.
- In these nodules the artery, which has now turned into an arteriole, occupies an eccentric position but is still called the central artery.

White Pulp

- During its passage through the white pulp, the artery also divides into numerous radial branches that supply the surrounding lymphoid tissue.
- Each lymphoid nodule is surrounded with a marginal zone consisting of many blood sinuses and loose lymphoid tissue. A few lymphocytes and many active macrophages are found there.
- The marginal zone contains an abundance of blood antigens and thus plays a major role in the immunological activities of the spleen.

White Pulp

- After leaving the white pulp, the sheath of lymphocytes slowly thins and the central artery (arteriole) subdivides to form straight **penicillar arterioles**.
- Near their termination, some of the penicillar arterioles are surrounded by a thick sheath of reticular cells, lymphoid cells, and macrophages (PAMS) and called sheathed arteriol.
- How the blood is delivered to the trabecular veins occurs in different ways explained by theories of splenic circulation.

Red Pulp

- **The red pulp** is composed of **splenic cords** and **sinusoids**.
- **The splenic cords** contain a network of reticular cells and reticular fibers.
- The splenic cords contain T and B lymphocytes, macrophages, plasma cells, and many blood cells (erythrocytes, platelets, and granulocytes).
- The splenic cords are separated by irregularly shaped wide sinusoids.

Red Pulp

- Elongated endothelial cells line the sinusoids of the spleen with the long axes parallel to the long axes of the sinusoids.
- These cells are enveloped in reticular fibers and surrounded with **incomplete basal lamina**.
- Because the spaces between the endothelial cells of the splenic sinusoids are 2–3 μm in diameter or smaller, **only flexible cells are able to pass** easily from the red pulp cords to the lumen of the sinusoids.

Functions of the Spleen

- **1. Filtration of the blood.** The cells of the spleen phagocytose, and produce immunological responses against blood-borne antigens.
- **2. The white pulp of the spleen is an important production site of lymphocytes.**
- **3. Destruction of aged Erythrocytes.**
- Erythrocytes have an average life span of around 120 days, after which they are destroyed, mainly in the spleen. A reduction in their flexibility and changes in their membrane seem to be the signals for their destruction. Degenerating erythrocytes are also removed in the bone marrow. Macrophages in the splenic cords engulf and digest the destroyed erythrocytes.

A man in a dark blue suit, light blue shirt, and dark tie is holding a large white rectangular sign. The sign has the words "Thank You" written on it in a large, bold, grey, sans-serif font. The man's hands are visible at the edges of the sign, holding it up. The background is a plain, light grey.

Thank You

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Thymic lobule

